

Estimating the uncertainty of using GPS radio occultation data for climate monitoring: Intercomparison of CHAMP refractivity climate records from 2002 to 2006 from different data centers

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[1] To examine the suitability of GPS radio occultation (RO) observations as a climate benchmark data set, this study aims at quantifying the structural uncertainty in GPS RO-derived vertical profiles of refractivity and measured refractivity trends obtained from atmospheric excess phase processing and inversion procedures. Five years (2002–2006) of monthly mean climatologies (MMC) of retrieved refractivity from the experiment aboard the German satellite CHAMP generated by four RO operational centers were compared. Results show that the absolute values of fractional refractivity anomalies among the centers are, in general, $\leq 0.2\%$ from 8 to 25 km altitude. The median absolute deviations among the centers are less than 0.2% globally. Because the differences in fractional refractivity produced by the four centers are, in general, unchanging with time, the uncertainty of the trend for fractional refractivity anomalies among centers is $\pm 0.04\%$ per 5 years globally. The primary cause of the trend uncertainty is due to different quality control methods used by the four centers, which yield different sampling errors for different centers. We used the National Centers for Environmental Prediction reanalysis in the same period to estimate sampling errors. After removing the sampling errors, the uncertainty of the trend for fractional refractivity anomalies among centers is between -0.03 and 0.01% per 5 years. Thus 0.03% per 5 years can be considered an upper bound in the processing scheme–induced uncertainty for global refractivity trend monitoring. Systematic errors common to all centers are not discussed in this article but are generally believed to be small.

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1. Introduction

[2] Long-term accurate measurements of atmospheric variations that are independent to data processing procedures are extremely important for climate monitoring. With a more complete spatial and temporal coverage than that from in situ observations, satellite measurements play an increasingly important role in global climate monitoring [Intergovernmental Panel on Climate Change, 2007]. Careful calibration procedures are necessary to determine trends in the climate system that are credible for societal objectives. Previous attempts at climate monitoring with in situ and remotely sensed data have suffered from inadequate calibration [Karl et al., 2006]. Thus, postprocessing of data to remove artifacts caused by inadequate calibration became necessary. However, because the intercalibration procedures used during periods of overlap are usually ill-determined and require subjective user judgment at some level,

independent efforts at creating climate data records using the same data sources produced significantly different estimates of trends of the climate system [National Research Council, 2004]. This has been the experience of data obtained by the microwave sounding units (MSU) on the NOAA TIROS Operational Vertical Sounders (TOVS) and the more recent Advanced Microwave Sounding Units (AMSU) [Christy et al., 2000; Mears et al., 2003; Grody et al., 2004; Zou et al., 2006; Karl et al., 2006] and the data contained in the historical radiosonde archive. See Karl et al. [2006] for a review.

[3] One proposed solution to this problem has been the creation and deployment of data sets whose calibration strategy is traceable to international standards (SI traceability) [Ohrring, 2007]. In the metrology of SI traceability, calibration is established through documented chains of calibration to the international standard defining the units of observation. With remote sensing systems, which cannot be retrieved periodically to check that calibration has been maintained, multiple calibration chains with independent physics are used to establish both calibration and the overall uncertainty in measurements. A time series of observations obtained using SI traceability can be considered calibrated with uncertainties determined by measurement and therefore requires no assumption of stability, no periods of overlap, and no need for adjustment of data. The data types that can be obtained in such a way are climate benchmarks. Two properties of climate benchmarks are that (1) contemporaneous data by different instruments agree to within stated uncertainties of the measurement, and (2) the trends in the observations as estimated by independent groups agree to within stated uncertainties of the measurement.

[4] Global Positioning System (GPS) radio occultation (RO), which is a technique with all-weather capability and high vertical resolution (from ~ 300 m near the surface to ~ 1.5 km at 40 km), has been proven to be very useful for weather prediction and global climate monitoring [Rocken et al., 1997; Wickert et al., 2001a; Kuo et al., 2004; Foelsche et al., 2008a, 2008b; Anthes et al., 2008; Ho et al., 2007; Ho et al., 2009a, 2009b]. GPS RO is the first satellite remote sensing technique where its fundamental observable, the delay of the occulted signal's phase due to the atmosphere and ionosphere, is accomplished via precise measurement of time that is traceable to ultrastable international standards (atomic clocks) on the ground. The traceability is accomplished by "simultaneous differencing," wherein the observing low Earth orbiting (LEO) satellite with the GPS receiver observes a GPS transmitter at zenith while simultaneously tracking another GPS satellite whose signal is occulted by the Earth's limb. The clocks of the GPS satellites are observed and subsequently corrected by stations on the ground that are themselves calibrated by the atomic clocks that define the international standard of time [Hardy et al., 1994]. This traceability makes GPS RO a strong candidate for a climate benchmark [Goody et al., 1998, 2002]. Property (1) listed above has already been tested for GPS RO: the mean dry temperature difference between the collocated soundings of the German CHAMP and Argentine SAC-C GPS RO missions [Hajj et al., 2004] and the collocated soundings of two receivers from the COSMIC constellation is within 0.1 K [Anthes et al., 2008; Foelsche et al., 2009; Ho et al., 2009a].

[5] We seek to test a climate benchmarking capability of GPS RO by assessing the degree to which observed trends agree among the independent processing groups, the property (2) listed above. While trends in the atmospheric phase delay as determined by different analysis groups should agree, it is not clear whether trends in any of the retrieved geophysical variables should agree. If the benchmarked observable is subjected to a consistent retrieval system for the entirety of its time series, we hypothesize that the systematic errors inherent to a retrieval system should not affect the *trend* in the retrieved quantity. With GPS RO, we choose to examine trends in vertical profiles of refractivity (N), the microwave index of refraction less unity in parts per million, as retrieved by the retrieval algorithms of four different GPS RO processing centers.

[6] Thorough theoretical error analyses have been performed for the GPS RO sounding technique [Kursinski et al., 1997; Rieder and Kirchengast, 2001; Steiner and Kirchengast, 2005], and we rely on such analyses to interpret the behavior of the retrieval errors associated with GPS RO in trending analysis. Several procedures are necessary to process RO phase delay into vertical profiles of refractivity. Those procedures expected to be the most likely sources of uncertainty in trend analysis are (1) precise orbit determination and clock synchronization to eliminate the effects of the geometric Doppler and of relative transmitter receiver oscillator drift, (2) the procedure to convert Doppler to bending angle, (3) the extrapolation of the ionospheric correction into the lower troposphere, necessary because of the influence of the ionosphere on measured phase delay of GPS signals, (4) the initialization of the Abelian integral transform which converts atmospheric bending angles to profiles of refractivity, and (5) the quality control (QC) algorithm used to distinguish acceptable data from unacceptable data (see section 2).

[7] The point of this paper is not to find a statistically significant climate trend, but to look for statistical significance, or absence thereof, of the *difference* in trends as determined by different independent retrieval centers. Should the difference in trends as determined by the centers be statistically indistinguishable from zero, then when a statistically significant *climate* trend does emerge in this data type, there will be no debate concerning errors in retrieval affecting the measured trend. It is a result of the basic observable—atmospheric phase delay—being traceable to the international definition of the second.

[8] The comparisons are based on statistics of the difference of trends in monthly mean refractivity climatology among four centers. The four GPS RO processing centers that contributed to this analysis are the GeoForschungsZentrum Potsdam (GFZ; German Research Centre for Geosciences), Germany, the NASA Jet Propulsion Laboratory (JPL), Pasadena, CA, USA, the University Corporation for Atmospheric Research (UCAR), Boulder, CO, USA, and the Wegener Center of the University of Graz (WegC), Graz, Austria. The common data set used here is that of the German CHAMP satellite mission [Reigber et al., 2005; Wickert et al., 2001a] between 2002 and 2006. We describe the inversion methods and quality control schemes used by the four centers in section 2. The preparation of CHAMP refractivity climatologies at GFZ, JPL, UCAR, and WegC is described in section 3. Monthly zonal average fractional

refractivity anomalies as generated by the four processing centers are compared in section 4. Time series comparisons of refractivity anomalies and trend analysis are in the same section. Possible causes for the refractivity differences among the four centers are discussed in section 5. We conclude our study in section 6.

2. Retrieval of CHAMP Refractivities at GeoForschungsZentrum Potsdam (GFZ), the Jet Propulsion Laboratory (JPL), University Corporation for Atmospheric Research (UCAR), and the Wegener Center (WegC)

[9] To obtain a refractivity profile for a GPS RO occultation event, one must perform (1) precise orbit determination (POD) and excess atmospheric phase processing, (2) a bending angle calculation, (3) an ionospheric correction, (4) an Abel integral inversion with upper boundary conditions, and (5) quality control (QC). We describe the data source used in this study in section 2.1. Starting in section 2.2, we detail the general approach of each processing step, and the assumptions and implementations used by each center for this particular step. A summary of retrieval details for each center is provided in Table 1. The general excess phase processing and refractivity inversion procedures and the sources of errors are described by *Kursinski et al.* [1997].

2.1. Data Sources

[10] Operational CHAMP refractivity profiles from GFZ (version 005) were downloaded from the Information System and Data Center (ISDC) (<http://isdc.gfz-potsdam.de/>). Recent information and related references on the operational standard and near-real time (NRT) orbit and occultation processing are given by *König et al.* [2006] and *Wickert et al.* [2005, 2009].

[11] CHAMP refractivity profiles processed by JPL were downloaded from the JPL Genesis website, <http://genesis.jpl.nasa.gov>. A description of JPL inversion algorithms can be found in [*Hajj et al.*, 2002]. Additions to the processing not covered by *Hajj et al.* [2002] include the use of the Canonical Transform (CT) technique for estimating bending angle and impact parameter from the Doppler shifts.

[12] Operational CHAMP refractivity profiles processed by UCAR (version 2007.3140) were downloaded from the UCAR COSMIC Data Analysis and Archive Center (CDAAC) (<http://cosmic-io.cosmic.ucar.edu/cdaac/index.html>). A description of UCAR inversion algorithms can be found online (see <http://cosmic-io.cosmic.ucar.edu/cdaac/doc/index.html>) and in the work of *Kuo et al.* [2004].

[13] The CHAMP refractivity profiles provided by WegC were also used in this study. WegC profile and climatology data are available from its global climate monitoring website, <http://www.wegcenter.at/globclim>. The Wegener Center's Occultation Processing System (OPS) is the retrieval component of the End-to-End Generic Occultation Performance Simulation and Processing System (EGOPS) [*Kirchengast et al.*, 2007]. In this study the version OPSv5.3 was used, an enhanced version of OPSv5.2 [*Foelsche et al.*, 2009] based on the heritage of the CHAMPCLIM retrieval scheme CCRv2.3 [*Gobiet and*

Kirchengast, 2004; *Borsche et al.*, 2006; *Foelsche et al.*, 2008a; *Gobiet et al.*, 2007; *Steiner et al.*, 2007].

2.2. Precise Orbit Determinations and Excess Atmospheric Phase Processing

[14] The first step in the retrieval process is to precisely determine the excess phase that accumulates in the GPS L1 and L2 phase measurements due to signal delay and bending in the Earth's atmosphere and ionosphere. This quantity is termed excess atmospheric phase. The raw GPS phase measurements include dominant effects due to GPS and CHAMP satellite motion, and the GPS transmitter and CHAMP receiver oscillator offsets. To generate excess atmospheric phase data, one needs to perform precise orbit determination (POD) of the GPS and CHAMP satellites. The quality of the CHAMP orbits is determined via analyses of orbit overlaps and Satellite Laser Ranging (SLR) residuals.

[15] Once the effect of satellite motion is removed, a double- or single-difference measurement technique is used to eliminate satellite clock errors and to derive the atmospheric excess phase of the occultation link. In the double-difference technique, the CHAMP 50 Hz L1 and L2 phase data from the occultation link, and an additional reference link are differenced with interpolated 1 Hz data from a fiducial global ground network [*Hajj et al.*, 2002; *Wickert et al.*, 2001b]. In the single-difference technique, the occultation and reference link data are differenced to remove the effect of the receiver clock errors, and solved-for high-rate GPS clock offsets are interpolated to remove the effects of the transmitter clock errors [*Wickert et al.*, 2002]. An ionospheric correction of L1 and L2 data on the reference link is sometimes performed with heavily smoothed L1–L2 to suppress the effect of LEO clock distribution errors and L2 phase noise [*Kuo et al.*, 2004; *Beyerle et al.*, 2005]. The calibrated atmospheric excess phase data including the GPS and CHAMP satellite orbit information are made available by the processing centers.

2.2.1. GFZ Procedure

[16] Precise orbits of the GPS and CHAMP satellites are generated using GFZ's precise orbit determination software EPOS-OC [*König et al.*, 2006] and provided to ISDC. The related orbit product, which is used for the standard occultation analysis is denoted as Rapid Science Orbit (RSO). Recent comparisons (2007) between CHAMP RSO data and SLR measurements yield a mean deviation of 5.9 cm in error estimate of altitude [*Wickert et al.*, 2009]. A double-difference technique is applied for the CHAMP standard processing to eliminate satellite clock errors and to derive the atmospheric excess phase of the occultation link [*Wickert et al.*, 2001b]. The ground data is provided by a fiducial global network of about 30 stations, operated jointly by JPL and GFZ. GFZ uses a smoothing L1–L2 technique to reduce the impact of the reference link L2 phase noise [*Beyerle et al.*, 2005].

2.2.2. JPL Procedure

[17] JPL uses the reduced-dynamic strategy and the GIPSY software package to determine the precise orbits of the GPS and CHAMP satellites [*Bertiger et al.*, 1994]. GPS satellite orbits and transmitter clock biases are based on the Fiducial Laboratories for International Natural Science Network solution generated at JPL, using global

Table 1. Overview of Different Implementations of Processing Steps of GFZ, JPL, UCAR, and WegC^a

URL/Processing Step	Implementations for Each Center
URL	
GFZ	http://isdg.gfz-potsdam.de
JPL	http://genesis.jpl.nasa.gov
UCAR	http://www.cosmic.ucar.edu
WegC	http://www.wegcenter.at/globclim
POD phase, orbit data	
GFZ	POD: EPOS-OC for RSO provision [König et al., 2006]; excess phase: double differencing, reference link smoothing
JPL	POD: reduced-dynamic strategy using GIPSY software [Bertiger et al., 1994]; excess phase: double differencing
UCAR	POD computed with Bernese 5.0 software [Dach et al., 2007]; excess phase: single differencing, reference link smoothing
WegC	GFZ orbit and phase data used
Bending angle calculation	
GFZ	FSI below 15 km [Jensen et al., 2003]; geometric optics used above 15 km
JPL	CT after Gorbunov [2002] applied to L1
UCAR	FSI [Jensen et al., 2003] applied to L1 in troposphere < dynamic L2 QC height; geometric optics used > dynamic L2 QC height
WegC	Geometric optics used for L1 and L2 BAs at all heights
Ionospheric correction	
GFZ	Linear combination of L1 and L2 BAs [Vorob'ev and Krasil'nikova, 1994]
JPL	Linear combination of L1 and L2 BAs [Vorob'ev and Krasil'nikova, 1994]; ionospheric correction term extrapolation below 10 km
UCAR	Linear combination of L1 and L2 BAs [Vorob'ev and Krasil'nikova, 1994]; ionospheric correction term extrapolation < dynamic L2 QC height
WegC	Linear combination of L1 and L2 BAs [Vorob'ev and Krasil'nikova, 1994]; ionospheric correction term extrapolation <15 km
Initialization of bending angles	
GFZ	Optimization after Sokolovskiy and Hunt [1996] with MSISE-90
JPL	Exponential function fit at 40–50 km and extrapolation above
UCAR	Optimization after Sokolovskiy and Hunt [1996] with fitting backgr. prof. (NCAR climatic extrapolation to 150 km), dynamic estimation of the top fit height, background and observed errors [Lohmann, 2005]
WegC	Statistical optimization >30 km with ECMWF short-term forecasts and MSISE-90 to 120 km [Healy, 2001], dynamic estimation of observed errors and inverse covariance weighting [Gobiet and Kirchengast, 2004; Gobiet et al., 2007]
Quality control	
GFZ	QC of forward differences of excess phases; QC of bending angles; QC of N using ECMWF analyses: >10% ΔN
JPL	QC of Doppler shift <6 km; QC of N , $T < 40$ km: ECMWF analyses, >10% ΔN and >10 K ΔT rejected
UCAR	QC of raw L1 Doppler (truncation); QC of L2 Doppler (reject if dynamic QC height >20 km); QC of bending angle (reject if top fit height <40 km); QC of N using climatology (reject if difference >50%)
WegC	QC of excess phases and bending angles; QC of N , T using ECMWF analyses: rejection if $\Delta N > 10\%$ in 5–35 km and/or $\Delta T > 20$ K in 8–25 km

^aAbbreviations are as follows: BA, bending angle; CT, canonical transform; ECMWF, European Centre for Medium-Range Weather Forecasts; FSI, full spectrum inversion; GFZ, GeoForschungsZentrum Potsdam; JPL, Jet Propulsion Laboratory; QC, quality control; UCAR, University Corporation for Atmospheric Research; WegC, Wegener Center.

ground tracking data [see Hajj et al., 2004]. Orbit precision is estimated based on consistency between daily solutions, using 5 hour overlap periods [Hajj et al., 2004]. Root Mean Square (RMS) velocity residuals are generally below 0.05 mm/sec. RMS ranging SLR residuals at the 10 cm level have been reported also. The transmitter and receiver clocks are calibrated using the double-differencing technique [Hajj et al., 2002]. Appropriate differences calculated between the four measurement links produces an atmospheric delay value that is quite insensitive to errors in the transmitter and receiver clocks. See also section 2.2.1 (GFZ procedure).

2.2.3. UCAR Procedure

[18] The CHAMP orbits and clocks are computed with the Bernese 5.0 GPS data processing package [Dach et al., 2007]. GPS satellite positions, velocities and clocks are known from the IGS Final results. A weighted least squares reduced-dynamic filtering approach is used to determine positions and velocities of the LEO [Svehla and Rothacher, 2003]. The L1 and L2 atmospheric excess phases are computed with a single-difference approach (occultation link – reference link) that removes the LEO clock offsets; the GPS satellite clock offsets are removed by interpolating

the 30 s IGS final clock products [Schreiner *et al.*, 2005]. The ionospheric correction of L1 and L2 on the reference link is performed by smoothing L1–L2 to suppress the effect of LEO clock distribution errors and L2 noise of the CHAMP GPS receiver [Kuo *et al.*, 2004; Beyerle *et al.*, 2005].

2.2.4. WegC Procedure

[19] CHAMP RO excess phase data and CHAMP orbital data provided by GFZ Potsdam (CH-AI-2-PD, version 2 data set) were used as inputs to the EGOPS OPSv5.3 processing.

2.3. Calculation of L1 and L2 Bending Angles

[20] The calculation of bending angles is generally divided into two height ranges: (1) above the lower troposphere where atmospheric multipath is not significant and (2) in the lower troposphere where atmospheric multipath occurs regularly due to gradients in moisture. In the upper height range, both the L1 and L2 bending angles are derived based on geometric optics from the time derivative of the excess phase (i.e., excess Doppler) after appropriate noise filtering [Vorob'ev and Krasil'nikova, 1994; Hajj *et al.*, 2002]. In the lower height range, radioholographic (wave optics) techniques, such as canonical transform (CT) and full-spectrum inversion (FSI), are employed that accurately transform the received L1 phase and amplitude data into impact parameter and bending angle even in strong atmospheric multipath environments [Gorbunov, 2002; Jensen *et al.*, 2003]. The transition height that divides the two height ranges is either fixed or determined dynamically based on the quality of the signals. The bending angles are calculated in the reference frame centered in the local center of sphericity of the reference ellipsoid under the estimated occultation point.

2.3.1. GFZ Procedure

[21] The GFZ calibrated atmospheric excess phase data, including the GPS and CHAMP satellite orbit information, are in use by several research groups without precise GPS processing capability (e.g., WegC, see section 2.2.4) as input for the application of inversion techniques to derive vertical atmospheric profiles. Excess phase noise filtering is applied with a Singular Value Decomposition fit [Press *et al.*, 1996], encompassing 70 points of the 50 Hz data. To correct for the effect of lower troposphere multipath, the FSI technique is applied below 15 km.

2.3.2. JPL Procedure

[22] JPL uses different methods to compute bending angle for the L1 and L2 frequencies. The CT technique is applied to the received phase and amplitude data for the L1 frequency, which is often available down to the surface. A non-CT version of L1 bending angles is also calculated in the same fashion as the L2 bending angles (non-CT) for use in the ionospheric correction (see 2.4.b below).

2.3.3. UCAR Procedure

[23] Fourier filtering is used to simultaneously smooth and differentiate the excess phase to obtain excess Doppler. The filter bandwidths used for L1 and L1–L2 are 2 Hz and 0.5 Hz, respectively. L1 bending angle in the lower troposphere is calculated by the FSI technique by use of geometric optical signal propagation from GPS and LEO orbits to closest circles. The FSI technique is applied up to the height where L2 data become unreliable. This transition height is determined dynamically for each occultation. For

the rest of the profile, L1 and L2 bending angles are calculated from the excess Doppler via geometric optics [Vorob'ev and Krasil'nikova, 1994].

2.3.4. WegC Procedure

[24] In the OPSv5.3 first an outlier rejection is performed on the 50 Hz sampling rate L1 and L2 phase delay profiles. Data points outside three times the standard deviation are substituted by the interval's mean of a 1 s running window. The phase delay profiles are smoothed by a regularization method after Syndergaard [1999] and then converted via numerical differentiation to Doppler shift profiles. Bending angles are computed based on geometric optics [e.g., Kursinski *et al.*, 1997].

2.4. Ionospheric Correction

[25] A linear combination of the L1 and L2 bending angle profiles is usually used for the ionospheric correction based on a procedure suggested by Vorob'ev and Krasil'nikova [1994]. The dual frequency correction is applied to the bending angles at the L1 and L2 frequencies, interpolated to a common impact parameter. In an effort to reduce the combined effect of L2 noise and ionospheric residuals, less heavily smoothed L1 is corrected by more heavily smoothed L1–L2 [Rocken *et al.*, 1997; Hajj *et al.*, 2002; Hocke *et al.*, 2003; Kuo *et al.*, 2004]. Owing to defocusing effects, low L2 transmit power, and incomplete knowledge of the P-code modulation, the L2 signal becomes too weak for robust tracking at lower altitudes. At these lower altitudes, L1–L2 is linearly extrapolated downward from higher altitudes to continue the ionospheric correction to the surface. A simulation study by Mannucci *et al.* [2006] suggests extrapolation may cause refractivity errors of ~0.05% near the upper altitude range where the L2 loss first occurs and the error declines rapidly below that height.

2.4.1. GFZ Procedure

[26] The linear combination of the L1 and L2 bending angle profiles suggested by Vorob'ev and Krasil'nikova [1994] is used for the ionospheric correction. Below 12 km altitude, L1–L2 is linearly extrapolated downward from higher altitudes to continue the ionosphere correction to the Earth's surface.

2.4.2. JPL Procedure

[27] The linear combination of the L1 and L2 bending angle profiles is used for the ionospheric correction based on a procedure suggested by Vorob'ev and Krasil'nikova [1994]. At altitudes below ~10 km, ionospheric L1–L2 is linearly extrapolated downward from higher altitudes to continue the ionospheric correction to the surface.

2.4.3. UCAR Procedure

[28] Ionospheric correction of the bending angles is performed by taking L1 and L2 bending angles at the same impact parameter [Vorob'ev and Krasil'nikova, 1994]. At altitudes below a height where the use of L2 data is stopped (this height, estimated individually for each occultation, is typically between 5 and 15 km), L1 is corrected by extrapolating L1–L2 linearly from above.

2.4.4. WegC Procedure

[29] The ionospheric contribution is corrected through linear combination of bending angles [Vorob'ev and Krasil'nikova, 1994]. Below 15 km impact height, ionospheric correction terms are linearly extrapolated downward from higher altitudes (L1–L2 differences at 15 to 25 km are

extended downward) to continue the ionospheric correction toward the surface.

2.5. Abel Integral Upper Boundary

[30] The upper limit of the Abel integral requires knowledge of the bending as a function of impact parameter up to the top of the atmosphere. Although the observational ionosphere-free bending angle is available from above 100 km to the surface, it is mostly noise at high altitudes. One approach used by three processing centers (GFZ, UCAR, WegC) to reduce the effect of error propagation downward from the upper stratosphere is to optimally mix the observational ionosphere-free bending angle with a background bending angle model [Sokolovskiy and Hunt, 1996]. The background bending angle profile can be derived from a climate model or from a combination of a climate and weather model; this profile can be either fixed or fitted to observations in some height interval. The weighting between observation and background may be either fixed or depend on the background errors and/or the observational errors estimated individually for each occultation. The Abel inversion is applied for the bending angles starting at high altitude [Fjeldbo et al., 1971].

2.5.1. GFZ Procedure

[31] The algorithm, described by Sokolovskiy and Hunt [1996], is applied for the optimization of the bending angles using the MSISE-90 climatology [Hedin, 1991].

2.5.2. JPL Procedure

[32] The JPL approach is to assume that the bending angle decays exponentially with height at high altitudes. Bending angle measurements within 40–50 km altitudes are fit to a simple exponential function. Bending angles above 50 km are extrapolated upward from the fitted exponential function.

2.5.3. UCAR Procedure

[33] The observational ionosphere-free bending angle is optimally mixed with the background bending angle model [Sokolovskiy and Hunt, 1996]. The background model is based on NCAR climatology exponentially extrapolated to 150 km [Randel et al., 2002] (section 4). This climatological bending angle is log linearly fitted to the observational bending angle between 20 km and a top fit height. The top fit height, which is restricted to 60 km, is determined as the maximum height where (1) the fractional difference between the observational bending angle and a smoothed one (with a 2 s window) does not exceed 60% and (2) the difference in slopes of logarithms of the observational and background bending angle profiles does not exceed 5%. Above 80 km there is smooth transition to climatological bending angle [Lohmann, 2007]. The weighting functions depend on the observational and background errors estimated individually for each occultation [Lohmann, 2005]. The observation error variance is estimated from the difference between the observation and the background between 60 and 80 km; the normalized background error variance is estimated between 20 km and the top fit height. The Abel inversion is applied for the optimized bending angle starting at 150 km by using a finite difference representation [Sokolovskiy et al., 2005].

2.5.4. WegC Procedure

[34] The retrieved bending angle profiles are combined with background information derived from European Centre

for Medium-Range Weather Forecasts (ECMWF) short-range forecast fields via statistical optimization with inverse covariance weighting [Healy, 2001; Rieder and Kirchengast, 2001; Gobiet and Kirchengast, 2004] above 30 km altitude. For each observed bending angle profile, a collocated refractivity profile is derived from the temporally closest forecast field available at four time layers (0000 UTC, 0600 UTC, 1200 UTC, and 1800 UTC, and 24 hour and 30 hour forecasts, respectively) at a resolution of $\sim 2.5^\circ \times 2.5^\circ$ (T42L60; increased to resolution T42L91 since February 2006). From the upper bound the ECMWF refractivity profile is expanded to 120 km using refractivity derived from the MSISE-90 climatology [Hedin, 1991], and transformed into a bending angle profile by a forward Abel transform. The error of the background profile is assumed to amount to 15% of the background bending angle with a vertical error correlation length of 9 km. The observation error is estimated from the variance of the observed profile between 65 km and 75 km (generally amounting to 1–4 μrad for CHAMP data at ~ 2 km vertical resolution), with a vertical error correlation length of 1.5 km assumed. The retrieval-to-background error ratio indicates an “effective” initialization top height of about 60 km to 77 km where the retrieved bending angle equals the background bending angle profile. The atmospheric profiles are background-dominated above the stratopause and observation-dominated below about 40 km [Gobiet and Kirchengast, 2004; Gobiet et al., 2007]. The Abel inversion, implemented as a numerical integration, is applied to the optimized bending angle profiles starting at 120 km.

2.6. Quality Control Methods

[35] The quality control (QC) methods used by the processing centers vary significantly, and they are applied at different processing stages. For example, the early stage QC methods examine the measured Signal-to-noise ratios (SNRs), excess phases, and excess Doppler shifts and modify or reject individual data points or entire profiles, while the final stage QC methods compare the retrieved refractivity profiles to either climatology or ECMWF analyses and discard the profiles with large differences.

2.6.1. GFZ Procedure

[36] A minimum of 650 connected data points must exist, for which the quotient of the forward differences of the excess phase L1 and L2 must be between 0.97 and 1.03 (no ionospheric disturbance) [see Beyerle et al., 2003]. This has the effect of eliminating occultations with evidence of strong ionospheric scintillation. Profiles are rejected for which the percentage of the “a priori” in the combination of measured and “background” bending angle for the optimization according to Sokolovskiy and Hunt [1996] exceeds 20% at 30 km. The fractional refractivity deviation from ECMWF for all altitudes must be within 10%.

2.6.2. JPL Procedure

[37] The Doppler-based quality control is imposed to eliminate data affected by poor phase-locked loop receiver tracking. It is particularly important to apply at lower altitudes (below about 6 km) but does not affect retrievals at the higher altitudes used in this comparison. After the retrievals are generated, the quality control is based on comparison to ECMWF analyses [Hajj et al., 2002]. The

following criteria are applied to the temperature and refractivity vertical profiles:

$$|N - N(\text{ECMWF})| < 0.1 N(\text{ECMWF}) \text{ over the whole profile below 40 km}$$

$$|T - T(\text{ECMWF})| < 10 \text{ K over the whole profile below 40 km.}$$

If either of these criteria is violated at any point in the profile the entire profile is removed from the data set. We note that JPL and WegC (see below) are two centers that use temperature as a quality control criterion. Implications of this choice are discussed below.

2.6.3. UCAR Procedure

[38] Data affected by receiver tracking errors are eliminated based on deviation (5–10 Hz) of raw L1 Doppler from a model based on orbits and CIRA + Q refractivity climatology [Kirchengast *et al.*, 1999]. An occultation is discarded if (1) the noise level on raw L2 or smoothed L1–L2 phase rates exceed 6 or 0.5 cm/sample above 20 km, respectively, (2) the top fit height for the background profile (see section 2.4.3) is less than 40 km, and (3) the retrieved refractivity differs fractionally from NCAR climatology by more than 50% between 10 and 40 km.

2.6.4. WegC Procedure

[39] The EGOPS OPS quality control (QC) includes internal (early stage) and external (final stage) QC. Internal QC is applied down to bending angle level and probes technical and consistency parts of the data and adjusts error estimates or rejects profiles during the retrieval as found needed. The external QC compares retrieved refractivity results to collocated ECMWF analysis profiles. Profiles featuring a relative refractivity difference to the ECMWF profiles >10% at any altitude level between 5 and 35 km, and/or a temperature difference >20 K between 8 and 25 km, are rejected. In total the QC removes about 25% of the CHAMP RO profiles entering the retrieval at excess phase level.

3. Preparation of Monthly Mean Climatologies of CHAMP Refractivity

[40] Monthly zonal mean climatologies (MMC) were obtained by “binning and averaging” of the retrieved refractivity profiles. For this study, zonal bins of 5° latitudinal width (i.e., 36 bins) were defined at a Mean Sea Level (MSL) altitude grid with regular 200 m spacing. CHAMP data in the period from January 2002 to December 2006 processed independently by the four centers are used to generate the MMCs. The binning and averaging procedures are performed independently by each center. Only refractivity profiles passing the differing QC tests implemented at each center are included in the binning procedure.

[41] Because refractivity decreases exponentially with height, it is easier to visualize refractivity anomalies among centers in a fractional sense. A fractional refractivity difference (i.e., $\Delta N/N$; hereafter we use ΔN to represent $\Delta N/N$) is related to a temperature difference (ΔT) at altitudes above 8 km globally (above 12 km in the tropics) where moisture has a negligible effect on refractivity (1% in fractional refractivity difference is $\sim 1\%$ fractional temperature difference which in turn is about a 2 K absolute temperature

difference). Because WegC performs geometric optics, and JPL and UCAR/GFZ perform CT and FSI in the lowest 8 km, respectively, to have a consistent comparison we present MMC ΔN comparisons in the height from 8 km to 30 km. We present results of the latitudinal and time series comparisons of the MMC fractional refractivity anomalies in sections 4.1 and 4.2, respectively. The trend analysis is presented in section 4.3.

4. Results

[42] In this section we present a comparison of climatological refractivity products generated by the four centers and then a comparison of their estimated trends. The source of uncertainty responsible for differences in climatologies of refractivity is dominantly retrieval error. We seek to find out to what extent retrieval error is systematic in time and can be expected to cancel in trend estimation.

4.1. Comparison of Zonal Average Refractivity Anomalies Among Four Centers

4.1.1. Zonal Average Refractivity Anomalies

[43] We generate zonal average fractional refractivity anomalies (in %) for each center using the following equation:

$$\Delta N(i, j) = (100\%/59) \times \sum_{k=1}^{k=59} \{MMC(i, j, k) - \overline{MMC}(i, j, k)\} / \overline{MMC}(i, j, k), \quad (1)$$

where i , j , and k are indices for monthly mean 5° latitude bins at a 200 m vertical grid from 8 km to 30 km for January 2002 to December 2006. Index i corresponds to the index of latitude bins, j to the index of altitude bins, and k to the month index. For the monthly mean, only those latitude, height, and month bins for all four centers containing more than five CHAMP profiles are included in the calculation. Because there are few CHAMP profiles in July 2006, MMCs of that month for the four centers are all set to zero. Therefore, 59 months of data are used here. $\overline{MMC}$ in equation (1) is the mean of MMCs from all four centers (intermodel mean). Figure 1 depicts the 59 month mean $\overline{MMC}$ from 8 km to 30 km. It shows that the mean refractivity decreases with altitude and is higher in the tropics and lower near the polar regions between 8 km to 20 km. This is qualitatively consistent with the global mean temperature of the same latitude-height domain (not shown).

[44] Figure 2 depicts the 2002–2006 mean zonal average refractivity according to each processing center with the intermodel mean subtracted (e.g., ΔN_{GFZ} , ΔN_{JPL} , ΔN_{UCAR} , and ΔN_{WegC}). Results show that although independent inversion algorithms and different quality control methods are used among centers, ΔN_{GFZ} , ΔN_{JPL} , ΔN_{UCAR} and ΔN_{WegC} agree to within $\pm 0.2\%$ except in the stratosphere. In the stratosphere above 25 km between 40° S and 40° N GFZ and JPL show a difference from the intermodel mean of approximately 0.3%. In the polar stratosphere, JPL and UCAR show a difference from the intermodel mean of up to 0.4%. Note that a difference for one center from an inter-center mean indicates that different retrieval and QC algorithms are being used, but does not provide direct

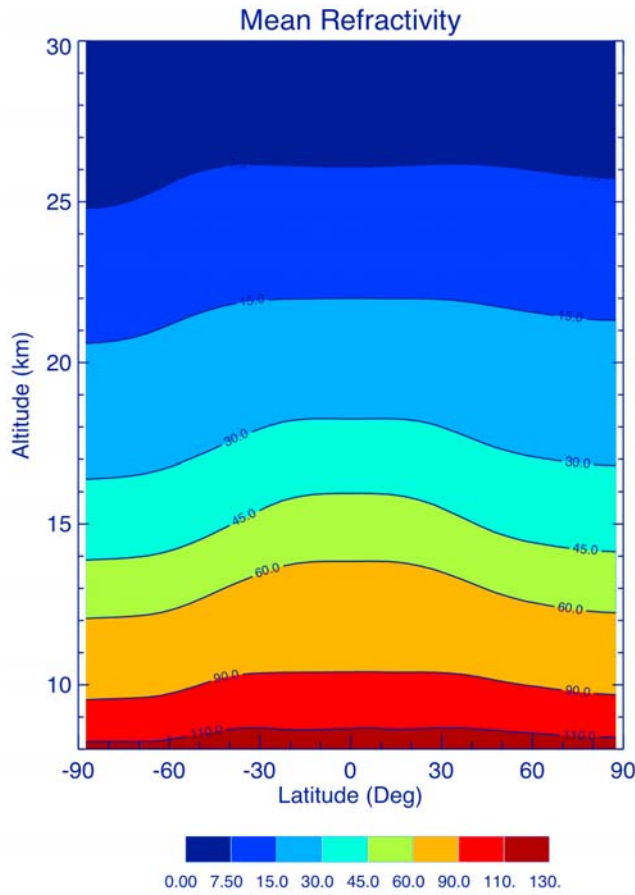


Figure 1. Mean monthly mean climatology (MMC) refractivity of GeoForschungsZentrum Potsdam (GFZ), the Jet Propulsion Laboratory (JPL), University Corporation for Atmospheric Research (UCAR), and the Wegener Center (WegC) from 8 to 30 km in the period from January 2002 to December 2006 (except for July 2006).

information on retrieval error. Intercenter anomalies are mainly caused by the different inversion algorithms used by the four centers. Differences in upper-altitude initialization of the Abel transform, that solves for refractivity given bending angle, is the most likely cause for low-latitude stratospheric differences. Differing quality control algorithms are likely responsible for both the polar stratospheric differences and the tropospheric differences between 40° and 50° in both hemispheres, which are particularly pronounced for UCAR (see below and section 5.1). In general, the interprocessing center retrieval difference is of the order of 0.2% in zonal average refractivity.

[45] The refractivity profile differences are mainly due to the combined effects of different initialization of the Abelian integral transform for refractivity and quality control (QC) methods. Differing QC methods adopted at each center imply that the times and locations of the profiles contributing to the MMCs differ among the centers. Comparing to ΔN_{JPL} , ΔN_{UCAR} has the opposite sign but with a similar magnitude from 60° S to 90° S and from 60° N to 90° N above 25 km, where the opposite sign anomaly is present between $\Delta N_{\text{JPL}}/\Delta N_{\text{UCAR}}$ and $\Delta N_{\text{GFZ}}/\Delta N_{\text{WegC}}$ near the tropics above 25 km. Because the four centers use

similar ionospheric correction schemes (although with different implementations), the different anomalies above 25 km are most likely owing to initialization with MSISE-90 for GFZ and exponential extrapolation for JPL. ΔN_{GFZ} has an opposite sign below 10 km with that of WegC. The mean fractional refractivity anomalies for GFZ, JPL, UCAR and WegC from 8 km to 30 km from 90° N to 90° S are equal to 0.04%, −0.01%, 0.04%, and −0.06%, respectively (Table 2). These relatively small anomalies among the four centers are encouraging and suggest that each center's processing and inversion algorithms are robust, despite distinct retrieval strategies and independent implementations, although the true atmospheric refractivity profiles are unknown.

4.1.2. Variation of Refractivity Anomalies

[46] To demonstrate the variation of refractivity anomalies, we also generated the Median Absolute Deviation (MAD, in %) for ΔN which is defined as

$$\Delta N^{\text{MAD}}(i, j) = (100\%/59) \times \sum_{k=1}^{k=59} | \text{MMC}(i, j, k) - \overline{\text{MMC}}(i, j, k) | / \overline{\text{MMC}}(i, j, k). \quad (2)$$

Figure 3 depicts two-dimensional distribution of the MAD of the fractional refractivity anomalies (e.g., $\Delta N_{\text{GFZ}}^{\text{MAD}}$, $\Delta N_{\text{JPL}}^{\text{MAD}}$, $\Delta N_{\text{UCAR}}^{\text{MAD}}$, and $\Delta N_{\text{WegC}}^{\text{MAD}}$) corresponding to Figure 2. It illustrates that even with different fractional refractivity anomalies among the four centers at different latitudes and heights, because GPS RO data are of very high precision (~ 0.1 K in temperature and $\sim 0.05\%$ in fractional refractivity) [e.g., Ho et al., 2009a], the $\Delta N_{\text{GFZ}}^{\text{MAD}}$, $\Delta N_{\text{JPL}}^{\text{MAD}}$, $\Delta N_{\text{UCAR}}^{\text{MAD}}$, and $\Delta N_{\text{WegC}}^{\text{MAD}}$ agree to within 0.1% for all four centers between 50° S to 50° N below 25 km. Below 25 km from 50° S to 90° S and from 50° N to 90° N, all $\Delta N_{\text{GFZ}}^{\text{MAD}}$, $\Delta N_{\text{JPL}}^{\text{MAD}}$, $\Delta N_{\text{UCAR}}^{\text{MAD}}$ and $\Delta N_{\text{WegC}}^{\text{MAD}}$ are within 0.2%, except for $\Delta N_{\text{JPL}}^{\text{MAD}}$ near 90° S near 10 km. Above 25 km, $\Delta N_{\text{UCAR}}^{\text{MAD}}$ has a similar pattern as that from $\Delta N_{\text{JPL}}^{\text{MAD}}$ especially over the polar regions, but with a smaller magnitude. $\Delta N_{\text{JPL}}^{\text{MAD}}$ is larger than 0.3% between 90° S to 60° S above 25 km, which may be due to the smaller number of profiles in the JPL data (see below).

4.1.3. Variation of Sample Numbers Among Four Centers

[47] The mean differences in Figure 2 and the MADs larger than 0.2% in Figure 3 especially over the polar regions above 25 km may be primarily due to the sampling errors among the different centers. Because different QC schemes are used at the four centers, the total sample numbers for each latitudinal bin for the different centers vary considerably. Figure 4 depicts the 59 month mean of the sample number for each 5° bin at the height of 25 km for the four centers, where the sample numbers at all other levels are close to that at 25 km. The very small sample number near the polar regions may be partly responsible for the large ΔN and ΔN^{MAD} (especially in case of JPL) in Figures 2 and 3 above 25 km in the same regions, respectively.

[48] Analysis of the data used in the JPL MMCs suggests that the lower sample number at high latitudes is due to JPL's QC criteria. JPL uses the most stringent QC criteria among the four centers for two reasons: (1) comparisons to

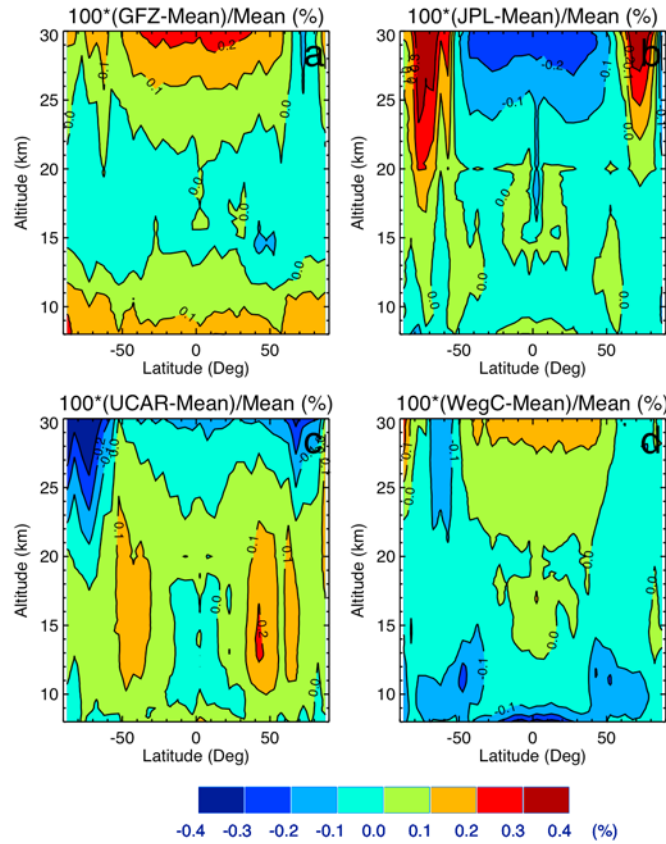


Figure 2. Difference in the mean zonal average refractivity over the period from January 2002 to December 2006 for each of the four contributing processing centers from the intermodel mean: (a) GFZ, (b) JPL, (c) UCAR, and (d) WegC.

analyses are applied at the highest altitude (40 km versus 35 km for WegC) and (2) JPL applies QC to temperature and refractivity, not refractivity alone. JPL analyzed the sample number passing QC at high latitudes and found a strong seasonal dependence. This suggests that the QC criteria are removing profiles based on geophysical conditions, primarily when strong temperature fluctuations exist in the high-latitude stratosphere. Sample number minima were found during winter conditions. We expect that future studies will be conducted with modified QC criteria to create more similarity among the profiles included in the comparisons.

[49] The larger sample number difference in southern midlatitudes between UCAR and the other three centers may partly be responsible for the $\Delta N_{\text{UCAR}} \sim 0.1\%$ in the same regions. Note that due to various QC schemes used by the four centers, it is not guaranteed that the same profiles are used by different centers for regions with similar sample number. This may explain part of the $\Delta N_{\text{UCAR}} \sim 0.2\%$ in the northern midlatitudes. This is confirmed by the sampling error analysis in section 5.1. The variation of sample numbers with times among the four centers also affects the comparison of time series refractivity anomalies (see the next section).

4.2. Time Series Comparison of Refractivity Anomalies

[50] To further quantify the time-dependent anomalies among centers and understand the causes of the differences,

we compute time series of fractional refractivity anomalies (in %) for each center (e.g., $\Delta N_{\text{GFZ}}^{\text{Time}}$, $\Delta N_{\text{JPL}}^{\text{Time}}$, $\Delta N_{\text{UCAR}}^{\text{Time}}$, and $\Delta N_{\text{WegC}}^{\text{Time}}$) at four vertical layers (l) including the 8–30 km layer ($l = 1$), the upper troposphere (the 8–12 km layer, $l = 2$), the lower stratosphere including tropopause (the 12–20 km layer, $l = 3$), and the midstratosphere (the 20–30 km layer, $l = 4$) at six latitudinal zones (n) from 90° N to 90° S ($n = 1$, global, actually from 87.5° N to 87.5° S), 90° N to 60° N ($n = 2$, northern high latitudes, actually from 87.5° N to 60° N), 60° N to 20° N ($n = 3$, northern midlatitudes and subtropics, further referred to as “midlatitude” for brevity), 20° N to 20° S ($n = 4$, tropics), 20° S to 60° S ($n = 5$, southern subtropics and midlatitudes, further referred to as “midlatitude” for brevity), and 60° S to 90° S ($n = 6$, southern high latitudes, actually from 60° S to 87.5° S). The equation we use is

$$\Delta N^{\text{Time}}(l, n, k) = 100\% \times \left\{ \frac{\text{MMC}(l, n, k) - \overline{\text{MMC}}(l, n, k)}{\overline{\text{MMC}}(l, n, k)} \right\} \quad (3)$$

where k is the index of the month bin ($k = 1$ to 60), and $\text{MMC}(l, n, k)$ is the mean MMC for four centers for each layer, zone, and month bin. Only those latitude, height, and month bins for all four centers containing more than five CHAMP profiles are included in the calculation. ΔN^{Time} for July 2006 is set to zero.

Table 2. Means and Standard Deviations of the Time Series of Fractional Refractivity Differences for GFZ, JPL, UCAR, and WegC to the Mean of All Four Centers for Six Latitudinal Zones and at Four Vertical Layers^a

Height Layers (km)	FR Mean (SD)/SE Mean (SD)			
	GFZ	JPL	UCAR	WegC
<i>90°N–90°S</i>				
8–30	0.04 (0.02)/0.06(0.02)	−0.01(0.03)/−0.03(0.02)	0.04(0.03)/0.02(0.02)	−0.06 (0.02)/−0.06(0.02)
8–12	0.1 (0.02)/0.12(0.02)	−0.02(0.04)/−0.02(0.02)	0.02 (0.03)/0.0(0.02)	−0.11 (0.03)/−0.11(0.03)
12–20	−0.02 (0.02)/0.0(0.02)	−0.01(0.04)/−0.02(0.03)	0.06 (0.03)/0.04(0.02)	−0.03 (0.02)/−0.02(0.03)
20–30	0.02 (0.03)/0.04(0.03)	−0.02(0.05)/−0.07(0.03)	0.01 (0.04)/0.01(0.03)	0.0 (0.03)/0.02(0.03)
<i>60°N–90°N</i>				
8–30	0.06 (0.08)/0.06(0.07)	−0.03 (0.12)/−0.01(0.08)	0.04 (0.08)/0.02(0.07)	−0.06 (0.08)/−0.06(0.12)
8–12	0.13 (0.1)/0.14(0.09)	−0.05 (0.15)/−0.01(0.09)	0.02 (0.08)/−0.01(0.05)	−0.1 (0.1)/−0.11(0.13)
12–20	−0.01(0.07)/0.0(0.08)	−0.04 (0.13)/−0.01(0.08)	0.07 (0.09)/0.04(0.07)	−0.02 (0.07)/−0.02(0.13)
20–30	−0.04 (0.17)/−0.14(0.16)	0.09 (0.28)/−0.02(0.16)	0.0 (0.13)/0.02(0.12)	−0.05 (0.14)/0.01(0.17)
<i>20°N–60°N</i>				
8–30	0.02 (0.05)/0.07(0.04)	−0.03 (0.07)/−0.04(0.03)	0.08 (0.06)/0.03(0.04)	−0.07 (0.04)/−0.06(0.05)
8–12	0.09 (0.05)/0.13(0.04)	−0.01 (0.08)/−0.03(0.03)	0.05 (0.06)/0.01(0.04)	−0.12 (0.05)/−0.11(0.05)
12–20	−0.05 (0.06)/0.0(0.05)	−0.03 (0.08)/−0.03(0.05)	0.12 (0.07)/0.06(0.05)	−0.05 (0.05)/−0.04(0.06)
20–30	0.02 (0.05)/0.04(0.05)	−0.07 (0.07)/−0.08(0.05)	0.04 (0.05)/0.02(0.04)	0.01 (0.05)/0.02(0.05)
<i>20°N–20°S</i>				
8–30	0.05 (0.02)/0.06(0.02)	−0.01 (0.03)/−0.03(0.03)	0.0 (0.02)/0.0(0.02)	−0.04 (0.01)/−0.04(0.02)
8–12	0.1 (0.02)/0.1(0.02)	0.0 (0.02)/−0.01(0.02)	0.01 (0.02)/0.01(0.02)	−0.1 (0.02)/−0.1(0.02)
12–20	0.0 (0.03)/0.01(0.03)	0.0 (0.05)/−0.02(0.05)	−0.01 (0.03)/0.0(0.03)	0.01 (0.02)/0.0(0.03)
20–30	0.05 (0.02)/0.05(0.02)	−0.09 (0.03)/−0.09(0.03)	0.0 (0.03)/−0.01(0.03)	0.04 (0.02)/0.04(0.02)
<i>20°S–60°S</i>				
8–30	0.03 (0.03)/0.06(0.03)	−0.02 (0.05)/−0.03(0.03)	0.06 (0.04)/0.04(0.04)	−0.08 (0.04)/−0.07(0.03)
8–12	0.09 (0.03)/0.11(0.03)	−0.01 (0.05)/−0.02(0.04)	0.03 (0.04)/0.01(0.03)	−0.11 (0.04)/−0.1(0.03)
12–20	−0.03 (0.04)/0.0(0.04)	0.0 (0.06)/−0.02(0.04)	0.09 (0.06)/0.07(0.05)	−0.06 (0.05)/−0.05(0.04)
20–30	0.03 (0.04)/0.04(0.04)	−0.08 (0.04)/−0.01(0.03)	0.05 (0.05)/−0.01(0.03)	0.01 (0.03)/0.02(0.03)
<i>60°S–90°S</i>				
8–30	0.05 (0.06)/0.07(0.05)	0.01(0.11)/−0.02(0.06)	0.01 (0.08)/0.0(0.05)	−0.08 (0.06)/−0.05(0.07)
8–12	0.13 (0.07)/0.13(0.06)	−0.03 (0.13)/−0.06(0.06)	0.01 (0.09)/−0.02(0.05)	−0.11 (0.07)/−0.1(0.08)
12–20	−0.02 (0.06)/0.0(0.05)	0.02(0.12)/−0.03(0.06)	0.04 (0.08)/0.04(0.05)	−0.04 (0.06)/−0.01(0.07)
20–30	0.0 (0.18)/0.05(0.13)	0.18(0.4)/0.0(0.17)	−0.1 (0.28)/−0.02(0.11)	−0.08 (0.11)/−0.03(0.09)

^aThe values of standard deviations of time series of fractional refractivity differences are shown in the parentheses. SE is for the mean and standard deviation of the time series of fractional refractivity differences after eliminating the sampling errors. Means and standard deviations are in percentages. Abbreviations are as follows: FR, fractional refractivity; GFZ, GeoForschungsZentrum Potsdam; JPL, Jet Propulsion Laboratory; UCAR, University Corporation for Atmospheric Research; WegC, Wegener Center.

[51] Figures 5 and 6 show the time series of monthly zonal average fractional refractivity anomalies with the interprocessing center mean subtracted, Figure 5 for the 12–20 km layer (the troposphere) and Figure 6 for the 20–30 km layer (the stratosphere). The resulting curves are the centers' monthly retrieval differences from the monthly intercenter mean. The fluctuations of intermonthly natural variability are removed as are retrieval differences systematic to all centers' retrieval algorithms. Two qualitative features can be inferred from these figures: (1) individual centers' anomalies that are persistent in time, and (2) individual centers' anomalies that show large intermonthly variance. For example, the global troposphere (Figure 5a) shows low intermonthly variance of anomalies but a clear persistent anomaly for just one center. On the other hand, the northern high-latitude troposphere (Figure 5b) shows large intermonthly variance in anomalies for all centers and no obvious persistent anomaly for any center. Note that in this type of plot, it is impossible for only one center to show large anomalies, persistent or varying. It is always necessary for the other centers to offset an individual center with exceptional behavior because the intermodel mean is subtracted. The persistent anomalies (mean) and anomaly variability (std) by latitude zone, height interval, and latitude bin, are given in Table 2.

[52] In the troposphere (Figure 5), the tropics show far less intermonthly variance of anomalies than do middle and

high latitudes since the tropical troposphere is characterized by far less synoptic variability in temperature and pressure. JPL, in particular, shows heightened anomaly variability (0.12%) in northern and southern high latitudes (Figures 5b and 5f); the other centers show less than 0.09%. If the four centers sample the troposphere differently by selecting different subsets of the raw GPS RO data in forming monthly averages, then the synoptic variability will leak into the anomaly variability apparent in Figure 5. The unusual sampling of JPL in the high-latitude regions (Figure 4) very likely explains JPL's heightened anomaly variability in comparison to those from the other centers in the high-latitude troposphere as related to sampling. Furthermore, even when the anomaly variability is smoothed, the anomalies of UCAR are persistently positive (+0.12% in northern midlatitudes, +0.09% in southern midlatitudes, +0.06% globally) with respect to the other centers' in the midlatitudes. The persistent (time-invariant) anomalies of other centers are always less than 0.05% otherwise. This cannot come about because of synoptic variability and instead may be a systematic difference due to UCAR's retrieval algorithm in the midlatitudes. In section 5.1 we will show that the large anomaly variability in the middle and high latitudes, for JPL and UCAR in particular, is mostly induced by sampling error by removing sampling error with the assistance of an atmospheric analysis.

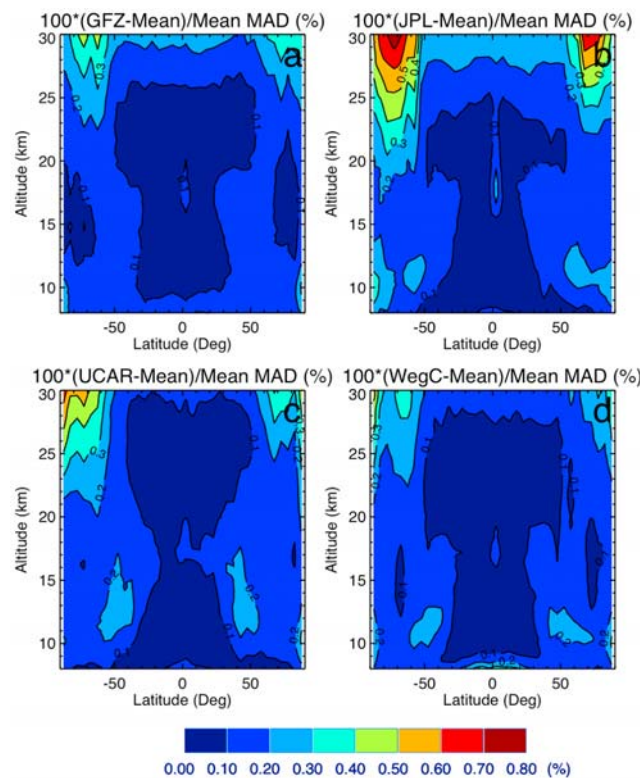


Figure 3. Median absolute deviation (MAD) of the fractional refractivity anomalies in the period from January 2002 to December 2006 to their mean from all four centers in the same period for (a) GFZ, (b) JPL, (c) UCAR, and (d) WegC.

[53] In the stratosphere (Figure 6), anomaly variability is extremely pronounced in northern and southern high latitudes and a persistent anomaly is shown for just one center in middle and low latitudes. Anomaly variability is strongly pronounced for JPL in the northern and southern high-latitude stratosphere, with standard deviations of 0.3% and 0.4%, respectively. The anomalies are especially pronounced in the northern hemispheric winter at high latitudes, when variations are $\approx 0.7\%$ and exceeding 1% on one occasion. Elsewhere JPL shows anomaly variability in the stratosphere of 0.03% to 0.08%. In the northern and southern high-latitude stratosphere, the other centers show anomaly variability of 0.03% to 0.08%. As above, we suspect that JPL's heightened anomaly variability in high latitudes is related to sampling error.

[54] Figure 7 shows that, in the months with larger differences in the number of RO soundings retained between JPL and UCAR (JPL has fewer samples), JPL shows the strongest positive anomalies (see Figure 6). Sample number differences alone will cause a center's MMC to differ from the intercenter mean because the atmosphere is not uniform within the 5° latitude bands. This is most apparent in the high latitudes and still noticeable in the southern midlatitude zone. The mean fractional refractivity anomaly for JPL varies from 0.09%, -0.07% , -0.09% , -0.08% , and 0.18% from the North to the South for each latitudinal zone (Table 2).

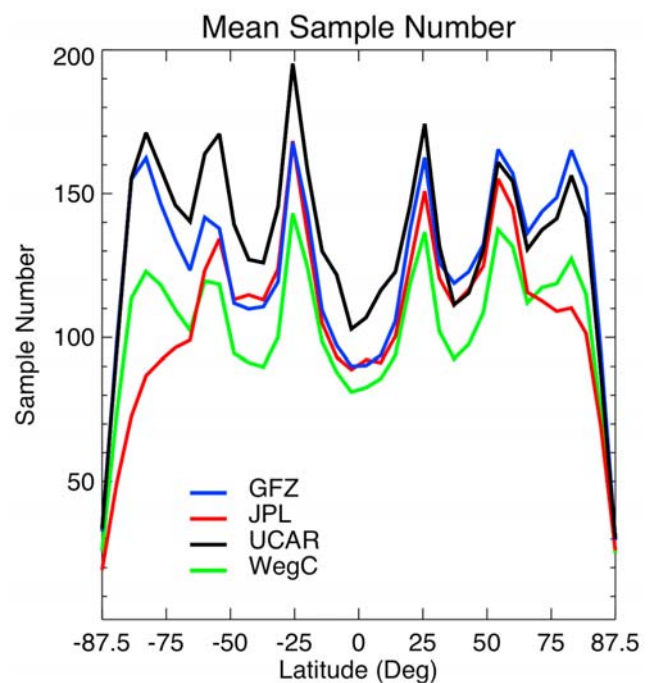


Figure 4. Monthly mean number of samples in latitudinal bins of 5° at 20 km altitude for GFZ, JPL, UCAR, and WegC in the period from January 2002 to December 2006.

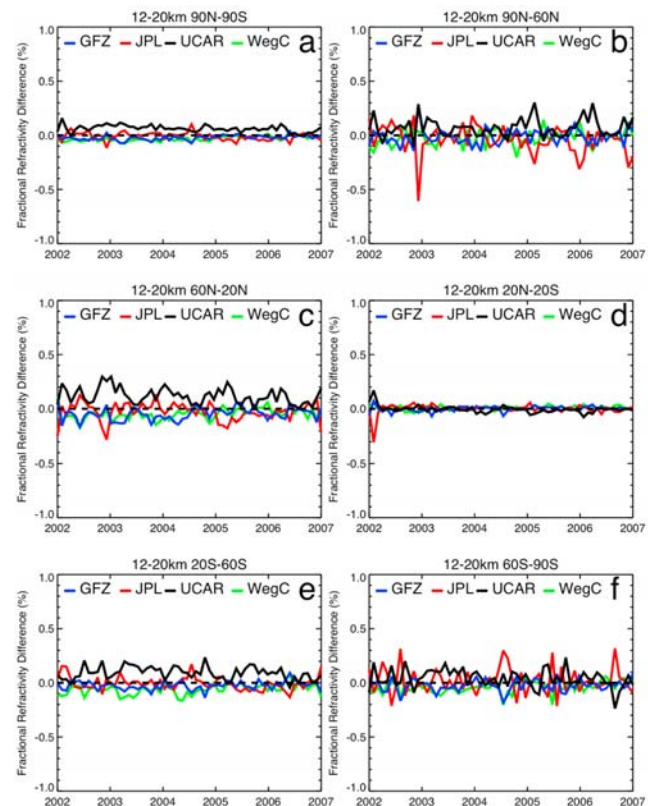


Figure 5. Time series of fractional refractivity anomalies among four centers for the 12–20 km layer for (a) the entire globe (90°N – 90°S), (b) the 90° – 60°N zone, (c) the 60° – 20°N zone, (d) the 20°N – 20°S zone, (e) the 20° – 60°S zone, and (f) the 60° – 90°S zone. The intermodel mean was subtracted on a monthly basis.

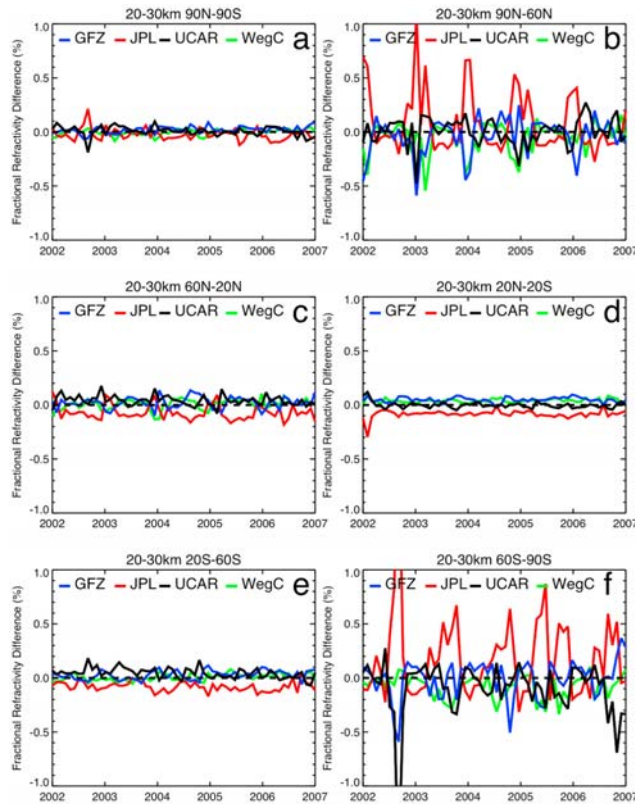


Figure 6. Same as Figure 5, but for the 20–30 km layer.

4.3. Trends and Variability in Refractivity Anomalies

4.3.1. Trend Analysis Method

[55] To quantify the uncertainty of using GPS RO data for climate monitoring, we examine the deseasonalized fractional refractivity anomalies among four centers (e.g., $\Delta N_{\text{GFZ}}^{\text{Deseason}}$, $\Delta N_{\text{JPL}}^{\text{Deseason}}$, $\Delta N_{\text{UCAR}}^{\text{Deseason}}$, and $\Delta N_{\text{WegC}}^{\text{Deseason}}$). Although trends evaluated here should not be considered indicative of any long-term climate trend as the period is clearly too short, the anomaly trend analysis is intended to show that even a short-term trend derived from GPS RO data is robust with regard to different processing implementations. If the robustness is true for short-term trends, then it can be expected true for long-term trends as well.

[56] The deseasonalized fractional refractivity anomaly is derived using the following equation:

$$\Delta N^{\text{Deseason}}(l, n, k) = 100\% \times \left\{ \text{MMC}(l, n, k) - \overline{\text{MMC}}(l, n, k') \right\} / \overline{\text{MMC}}^{\text{all}}(l, n, k'), \quad (4)$$

where k and k' are the indices of the month bin for the whole time series ($k = 1$ to 60) and of the month bin for the year ($k' = 1$ to 12). $\text{MMC}(l, n, k')$ is the mean MMC for each center for each layer (l), zone (n), and averaged over all years for a particular month (k'). $\overline{\text{MMC}}^{\text{all}}(l, n, k')$ is the mean MMC for all four centers for each layer, zone, and month of the year bin. Since the MMC for July 2006 is not available, $\overline{\text{MMC}}(l, n, k')$ and $\overline{\text{MMC}}^{\text{all}}(l, n, k')$ are constructed for the period from January 2002 to December 2005. The slope of the best linear fit of the deseasonalized fractional

refractivity anomalies for each center is defined as the trend (in %/5 years). The root mean square (RMS) difference between $\Delta N_{\text{GFZ}}^{\text{Deseason}}$ and the mean of $\Delta N_{\text{GFZ}}^{\text{Deseason}}$, $\Delta N_{\text{JPL}}^{\text{Deseason}}$, $\Delta N_{\text{UCAR}}^{\text{Deseason}}$, and $\Delta N_{\text{WegC}}^{\text{Deseason}}$ for each center (RMS_{GFZ} , RMS_{JPL} , RMS_{UCAR} , and RMS_{WegC}) are also computed. The mean trend of all four centers, the trend difference to the mean trend, and RMS differences for each center at each latitudinal zone and vertical layer are summarized in Table 3 and Table 4, respectively.

4.3.2. Trend Analysis of Fractional Refractivity Anomalies

[57] The trend of fractional refractivity anomalies represents variation of atmospheric density with time in the 8–30 km layer. Figure 8 depicts $\Delta N_{\text{GFZ}}^{\text{Deseason}}$, $\Delta N_{\text{JPL}}^{\text{Deseason}}$, $\Delta N_{\text{UCAR}}^{\text{Deseason}}$, and $\Delta N_{\text{WegC}}^{\text{Deseason}}$ for the southern high latitudes, tropics, and northern midlatitude zones for the 12–20 km layer (left panels), respectively. The fractional refractivity anomalies for other latitudinal zones are similar to those in Figure 8, and are not shown here. The best fit linear trend (the slope of the linear fit) of each processing center is also generated. The trend produced by each center individually is statistically insignificant because of the presence of natural variability and the brevity of the time series. Our interest, though, is in the difference between trends produced by the various centers, which can be statistically far more significant because natural variability is manifest in the various centers' products in exactly the same way and

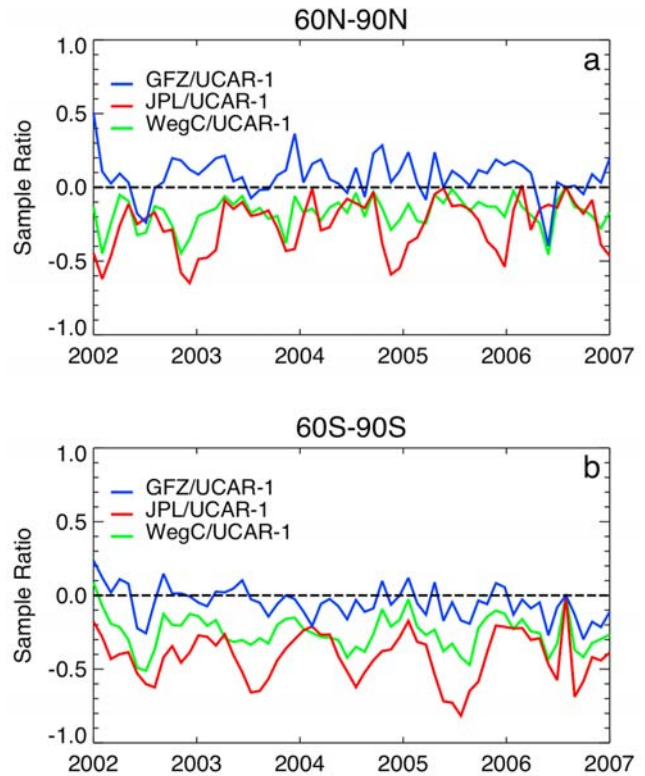


Figure 7. Sampling number ratio between GFZ and UCAR (defined as (GFZ Num – UCAR Num)/UCAR Num and written as (GFZ/UCAR – 1)), JPL and UCAR, and WegC and UCAR for (a) the 60°–90°N zone and (b) the 60°–90°S zone.

Table 3. Mean Trends for the Period 2002–2006 of Deseasonalized Fractional Refractivity Anomalies for GFZ, JPL, UCAR, and WegC and the GFZ Mean Anomalies of Four Centers, JPL Mean Anomalies of Four Centers, UCAR Mean Anomalies of Four Centers, and WegC Mean Anomalies of Four Centers for the Global (90°N–90°S) and Five Latitudinal Zones at Four Vertical Layers^a

Height Layers (km)	FR/SE (%/5 years)				
	Trend	GFZ Mean	JPL Mean	UCAR Mean	WegC Mean
<i>90°N–90°S</i>					
8–30	−0.03/−0.04	0.02/−0.01	−0.04/−0.01	−0.03/−0.01	0.04/0.03
8–12	0.13/0.11	0.01/−0.01	−0.04/−0.01	−0.02/0.0	0.04/0.03
12–20	−0.07/−0.1	0.03/−0.02	−0.03/0.01	−0.04/−0.01	0.05/0.02
20–30	−0.5/−0.46	0.05/0.02	−0.05/−0.03	−0.03/−0.04	0.04/0.05
<i>60°N–90°N</i>					
8–30	0.21/0.26	0.05/0.04	−0.12/−0.06	0.02/−0.03	0.05/0.06
8–12	0.33/0.35	0.05/0.05	−0.12/−0.08	0.01/−0.04	0.05/0.07
12–20	0.18/0.23	0.03/0.03	−0.1/−0.03	0.02/−0.03	0.05/0.03
20–30	−0.14/0.0	0.1/0.04	−0.2/−0.09	0.02/−0.05	0.09/0.1
<i>20°N–60°N</i>					
8–30	0.18/0.17	0.03/−0.03	−0.04/−0.01	−0.05/−0.01	0.06/0.06
8–12	0.23/0.22	0.02/−0.04	−0.04/−0.02	−0.04/−0.02	0.07/0.07
12–20	0.17/0.14	0.04/−0.03	−0.03/0.0	−0.06/−0.01	0.06/0.04
20–30	0.04/0.06	0.02/0.01	−0.04/−0.04	−0.04/−0.02	0.06/0.06
<i>20°N–20°S</i>					
8–30	0.0/−0.043	0.01/−0.02	0.0/0.0	−0.02/−0.01	0.02/0.0
8–12	0.02/−0.01	0.0/−0.02	−0.02/0.0	−0.01/0.0	0.03/0.01
12–20	−0.04/−0.06	0.02/−0.02	0.0/0.04	−0.04/−0.02	0.02/0.0
20–30	0.03/0.02	0.0/−0.01	0.01/0.02	−0.02/−0.02	0.01/0.01
<i>20°S–60°S</i>					
8–30	0.14/0.07	0.01/−0.03	−0.02/0.02	−0.03/0.01	0.04/−0.01
8–12	0.1/0.04	−0.01/−0.03	−0.01/0.04	−0.01/0.01	0.02/−0.02
12–20	0.13/0.03	0.02/−0.03	−0.04/0.02	−0.04/0.02	0.05/−0.01
20–30	0.34/0.32	0.05/0.01	−0.03/−0.01	−0.07/−0.04	0.05/0.04
<i>60°S–90°S</i>					
8–30	−0.34/−0.31	0.02/0.01	−0.02/−0.06	−0.05/0.0	0.05/0.05
8–12	0.31/0.31	0.0/−0.01	−0.02/−0.06	−0.04/0.03	0.06/0.05
12–20	−0.29/−0.24	0.02/0.01	−0.01/−0.06	−0.07/−0.02	0.06/0.07
20–30	−3.1/−3.0	0.13/0.12	−0.08/−0.07	−0.03/−0.07	−0.02/0.03

^aThe corresponding trend and trend difference to the mean trend for deseasonalized fractional refractivity anomalies are designated as “FR,” and the trend and trend difference to the mean trend for deseasonalized fractional refractivity anomalies after eliminating the sampling errors are designated as “SE.” Abbreviations are as follows: GFZ, GeoForschungsZentrum Potsdam; JPL, Jet Propulsion Laboratory; UCAR, University Corporation for Atmospheric Research; WegC, Wegener Center.

thus is completely removed in differencing the centers' trends.

[58] The left panels of Figure 8 (for the 12–20 km layer) show that the time series of the fractional refractivity anomalies of the four centers vary with different latitudinal zones (Figures 5 and 6). Nevertheless, the trends in $\Delta N_{\text{GFZ}}^{\text{Deseason}}$, $\Delta N_{\text{JPL}}^{\text{Deseason}}$, $\Delta N_{\text{UCAR}}^{\text{Deseason}}$, and $\Delta N_{\text{WegC}}^{\text{Deseason}}$ agree within ± 0.05 (%/5 years) globally (see Table 3), because their differences are nearly constant in time (with standard deviation–SD within 0.4%, see Table 2). The regional trends and the standard deviations in the trend estimates in the multicenter ensembles in the northern high-latitude, northern midlatitude, tropics, southern midlatitude, and southern high-latitude upper troposphere are 0.18 ± 0.04 , 0.17 ± 0.05 , -0.04 ± 0.02 , 0.13 ± 0.04 , and -0.29 ± 0.04 %/5 years, respectively (Table 3). The error bar is less on the trend variation (the standard deviation in the trend estimates) than on the mean trend, except in the tropical band, because the natural variability anomalies are common to all of the time series.

[59] The $\Delta N_{\text{GFZ}}^{\text{Deseason}}$, $\Delta N_{\text{JPL}}^{\text{Deseason}}$, $\Delta N_{\text{UCAR}}^{\text{Deseason}}$, and $\Delta N_{\text{WegC}}^{\text{Deseason}}$ for the southern high-latitude, tropics, and northern midlatitude zones for the 20–30 km layer are also plotted in Figure 8 (right), respectively. The seasonal fractional refractivity anomalies for JPL (also for UCAR but

smaller) in the southern high-latitude zone (also in northern high-latitude zone, not shown) in the 20–30 km layer are largely removed as a result of subtracting their seasonal cycle.

[60] Furthermore, we generate the differences between $\Delta N_{\text{GFZ}}^{\text{Deseason}}$, $\Delta N_{\text{JPL}}^{\text{Deseason}}$, $\Delta N_{\text{UCAR}}^{\text{Deseason}}$, and $\Delta N_{\text{WegC}}^{\text{Deseason}}$ and the mean deseasonalized fractional refractivity anomalies for all centers for the southern high-latitude, tropics, and northern midlatitude zones for the 12–20 km layer (Figure 9, left). These panels show that the differences of the deseasonalized fractional refractivity anomalies from the mean for the four centers are in general less than 0.2% for the 12–20 km layer, except for JPL during the summer months in the southern high-latitude zone. The peaks and valleys of the JPL and UCAR differences in the southern high-latitude zone correspond to the refractivity anomalies in Figure 5 in the same zone. The global mean RMS_{GFZ} , RMS_{JPL} , RMS_{UCAR} , and RMS_{WegC} are all between 0.02% and 0.03% for the 8–12 km layer and 12–20 km layer (Table 4). Both interannual variability of refractivity and random retrieval error contribute to these RMSs. The RMS_{JPL} is usually the largest among four centers especially at the high latitudes.

[61] Like those for the 12–20 km layer, the time series of $\Delta N_{\text{GFZ}}^{\text{Deseason}}$, $\Delta N_{\text{JPL}}^{\text{Deseason}}$, $\Delta N_{\text{UCAR}}^{\text{Deseason}}$, and $\Delta N_{\text{WegC}}^{\text{Deseason}}$ are very consistent with each other in the 20–30 km layer

Table 4. Same as Table 3, but for the Mean Root Mean Square Differences

Height Layer (km)	FR/SE (%/5 years)			
	GFZ–Mean	JPL–Mean	UCAR–Mean	WegC–Mean
<i>90°N–90°S</i>				
8–30	0.02/0.02	0.03/0.02	0.02/0.02	0.02/0.02
8–12	0.02/0.02	0.03/0.02	0.03/0.02	0.02/0.03
12–20	0.02/0.02	0.03/0.03	0.03/0.02	0.02/0.02
20–30	0.05/0.02	0.05/0.03	0.03/0.02	0.04/0.03
<i>60°N–90°N</i>				
8–30	0.07/0.06	0.11/0.07	0.07/0.07	0.07/0.1
8–12	0.09/0.07	0.12/0.09	0.07/0.07	0.08/0.12
12–20	0.07/0.06	0.1/0.07	0.08/0.06	0.06/0.11
20–30	0.11/0.12	0.16/0.11	0.11/0.11	0.1/0.14
<i>20°N–60°N</i>				
8–30	0.04/0.03	0.06/0.03	0.04/0.03	0.04/0.04
8–12	0.05/0.04	0.07/0.03	0.05/0.03	0.05/0.05
12–20	0.05/0.04	0.07/0.04	0.06/0.05	0.05/0.05
20–30	0.05/0.04	0.05/0.05	0.04/0.03	0.04/0.05
<i>20°N–20°S</i>				
8–30	0.02/0.02	0.03/0.03	0.02/0.02	0.01/0.01
8–12	0.02/0.02	0.02/0.02	0.02/0.02	0.02/0.02
12–20	0.02/0.03	0.04/0.05	0.03/0.03	0.02/0.02
20–30	0.02/0.03	0.03/0.03	0.03/0.03	0.01/0.02
<i>20°S–60°S</i>				
8–30	0.03/0.03	0.04/0.03	0.04/0.03	0.04/0.01
8–12	0.03/0.03	0.04/0.03	0.04/0.03	0.04/0.02
12–20	0.04/0.04	0.06/0.04	0.06/0.05	0.05/0.01
20–30	0.03/0.03	0.04/0.03	0.05/0.04	0.03/0.04
<i>60°S–90°S</i>				
8–30	0.05/0.04	0.1/0.05	0.07/0.04	0.05/0.06
8–12	0.06/0.05	0.12/0.06	0.08/0.05	0.06/0.07
12–20	0.05/0.04	0.1/0.06	0.08/0.05	0.05/0.07
20–30	0.14/0.11	0.28/0.12	0.2/0.09	0.09/0.09

(Figure 9, right). The mean trends for each zone from the North to the South are -0.14 , 0.04 , 0.03 , 0.34 , and $-3.1\%/5$ years (Table 3), respectively, where the standard deviation in the trend estimates for each center from the mean trend for the same zone are $\pm 0.1\%$, $\pm 0.04\%$, $\pm 0.01\%$, $\pm 0.05\%$, and $\pm 0.06\%/5$ years, respectively (Table 3). The RMS for the Tropical zone and midlatitudes are around 0.03 – 0.05% for all four centers (Table 4). The remaining episodic fractional refractivity anomalies in the southern high-latitude zone are still obvious after removing the seasonal cycle. This is very likely caused by the different subsets of profiles used by the four centers in forming the MMCs. These remaining anomalies have obvious impacts on trend analysis. Thus the MMC sampling errors among the four centers are evaluated in section 5 and are used to derive updated trends of the fractional refractivity anomalies among four centers, with sampling error estimates subtracted.

[62] Globally (8–30 km layer, 90° N to 90° S zone), the trends from each of the four centers differ from the mean trend over all centers by less than $\pm 0.04\%/5$ years. These differences are due to differences in implementation of the GPS RO retrieval chain (including different QCs) by different processing centers. Over all heights (8–30 km) the RMS stays between 0.02 and $0.03\%/5$ years for all centers for the globe (Table 4). This is an encouraging result suggesting significant value of GPS RO for global trend monitoring. Inasmuch as the naturally occurring interannual variability of annual average refractivity is approximately

$0.01\%/5$ years in the troposphere, some accounting for GPS RO retrieval error is still necessary (see below).

5. Discussion

5.1. The Uncertainty of Anomaly Trends due to the Sampling Errors Among Data Sets

[63] With different quality control schemes among the four centers, the CHAMP profiles contributing to each MMC bin vary in time and location [von Engel, 2006], leading to MMC differences among the four centers that noticeably impact our trend analysis. To reduce these sampling error effects on the trend analysis, we construct the improved MMCs for each center by subtracting estimates of the sampling errors from the respective MMC (similar to Pirscher et al. [2007] and Foelsche et al. [2008a]). The sampling errors are estimated using daily National Centers for Environmental Prediction (NCEP) reanalysis (on a 6 hour and 1° bin) from January 2002 to December 2006. The following procedure is used.

[64] 1. Interpolating N profiles from NCEP reanalysis (NCEP_{prof} hereafter) to the times and locations of each

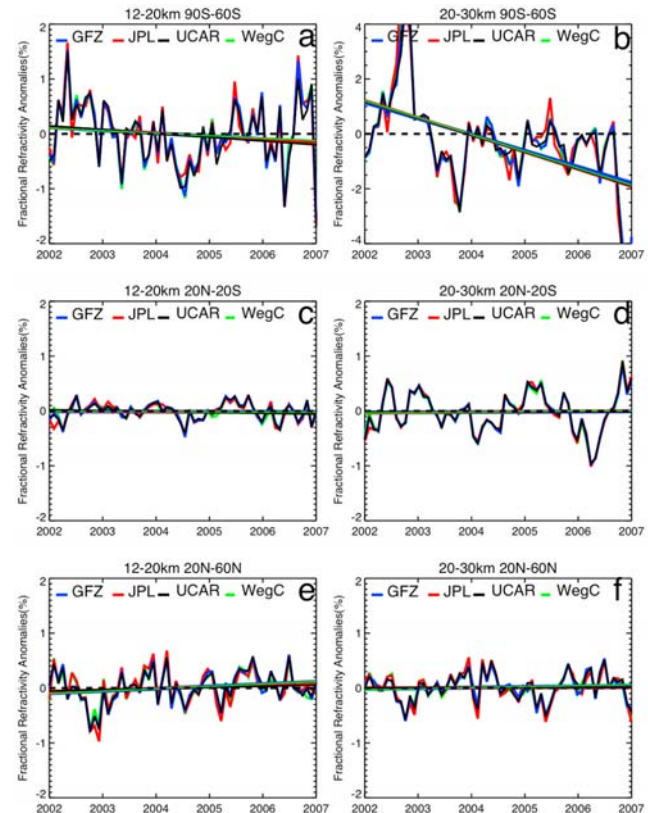


Figure 8. Deseasonalized fractional refractivity anomalies for each center in the 12–20 km layer for (a) the 60° – 90° S zone, (c) the 20° N– 20° S zone, and (e) the 20° – 60° N zone, and in the 20–30 km layer for (b) the 60° – 90° S zone, (d) the 20° N– 20° S zone, and (f) the 20° – 60° N zone. The 5 year trend for each center is shown as well. Note that Figure 8b has its ordinate range enlarged to $\pm 4\%$ (relative to $\pm 2\%$ of Figures 8a and 8c–8f).

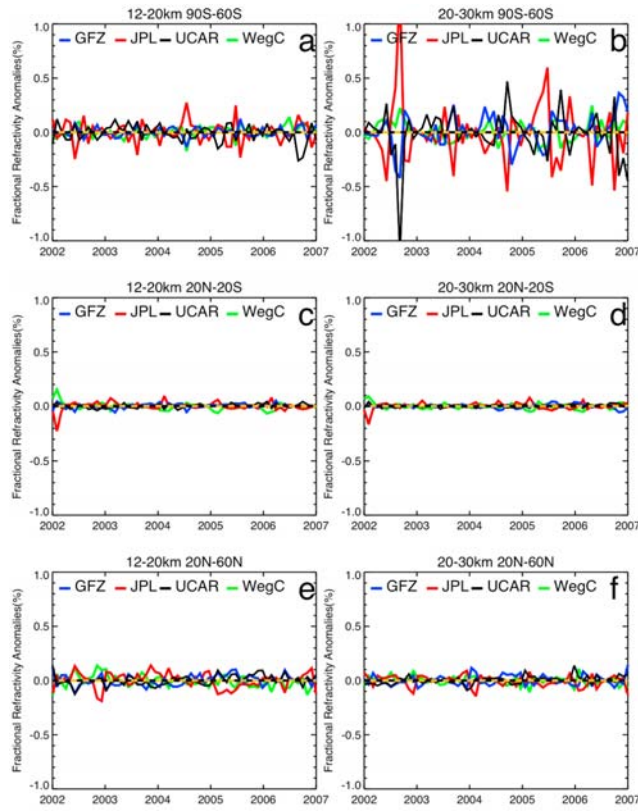


Figure 9. Differences between deseasonalized fractional refractivity anomalies for each center to their mean of all four centers in the 12–20 km layer in (a) the 60°–90°S zone, (c) the 20°N–20°S zone, and (e) the 20°–60°N zone, and in the 20–30 km layer in (b) the 60°–90°S zone, (d) the 20°N–20°S zone, and (f) the 20°–60°N zone. Note that Figure 9 is different from Figure 5, where Figure 5 is for the time series of fractional refractivity anomalies among the four centers (the seasonal variation is not subtracted).

CHAMP profile for different centers ($NCEP_{Int_prof}$), the following formula is used:

$$NCEP_{Int_prof} = \text{Interpolate}(NCEP_{prof}(\text{Time}_{CHAMP}, \text{Location}_{CHAMP})), \quad (5)$$

where $NCEP_{prof}$ refractivity is constructed using $NCEP$ temperature, moisture and pressure profiles, Time_{CHAMP} and Location_{CHAMP} are the time and location of each CHAMP profile. The “Interpolate” function in equation (5) represents a series of UCAR CDAAC operational temporal and spatial interpolation procedures.

[65] 2. Binning individual $NCEP_{Int_prof}$ and $NCEP_{prof}$ profile for different centers into the monthly mean climatologies as described in section 3, respectively. The binned $NCEP_{Int_prof}$ and $NCEP_{prof}$ are denoted as $NCEP_{Int_prof}(i, j, k)$ and $NCEP_{prof}(i, j, k)$, where i, j , and k are indices for the 5° latitude bin, 200 m vertical grid from 8 km to 30 km, and month bin from January 2002 to December 2006, respectively.

[66] 3. The sampling error of MMCs (MSE) is defined as

$$MSE(i, j, k) = \overline{NCEP_{Int_prof}(i, j, k)} - \overline{NCEP_{prof}(i, j, k)}. \quad (6)$$

The new MMCs (denoted as MMC^{NEW}) are then defined as

$$MMC^{NEW}(i, j, k) = MMC(i, j, k) - MSE(i, j, k). \quad (7)$$

Using the new MMCs for the four centers, we repeat the analysis of the time series comparison of refractivity anomalies and trend analysis in section 4. In adopting this approach, we have only assumed that the NCEP Reanalysis reproduces synoptic variability, the dominant cause of sampling error in CHAMP data, but not necessarily the interannual trend in the climate. The results will not depend on interannual trends in the NCEP Reanalysis. Findings of the new results are as follows.

[67] 1. Results demonstrate that both systematic refractivity anomalies and their variability decrease among the four centers, after reducing the impact of using distinct sets of profiles in the formation of the MMCs (varying quality control criteria). Figures 10 and 11 show the time series of the new monthly zonal average fractional refractivity anomalies with the interprocessing center mean subtracted, Figure 10 for the 12–20 km layer, and Figure 11 for the 20–30 km layer. The fractional refractivity anomalies of UCAR in the 12–20 km layer in the midlatitudes (see Figure 5 and Table 2) decrease from 0.12% to 0.06% in the northern midlatitude zone and from 0.09% to 0.07% in the

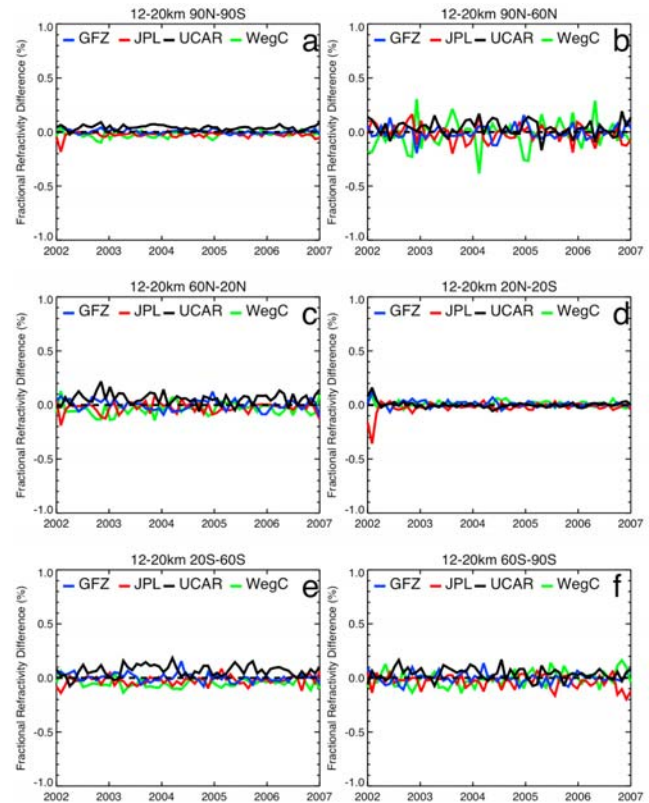


Figure 10. Same as Figure 5 (12–20 km layer, selected latitude bands), but showing the time series of refractivity anomalies with the sampling error subtracted.

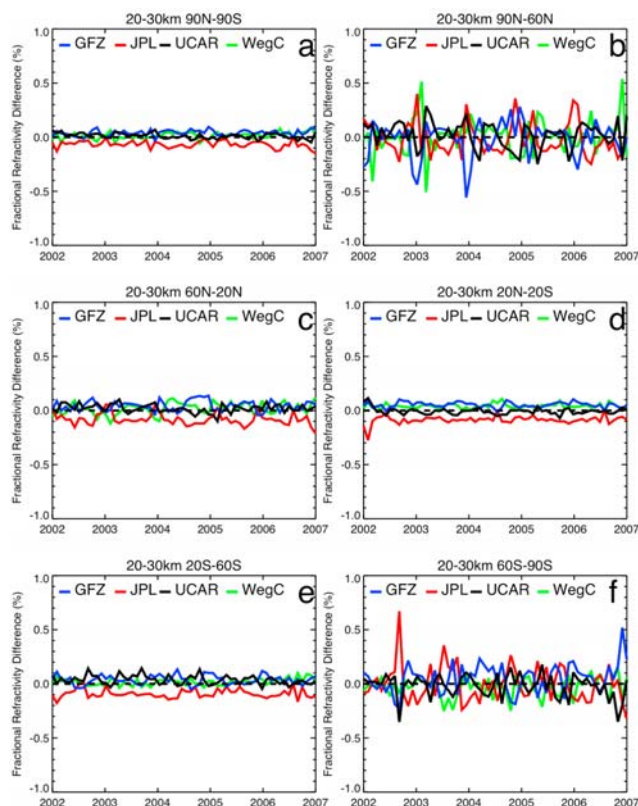


Figure 11. Same as Figure 10, but for the 20–30 km layer.

southern midlatitude zone, which is more consistent with the other three centers. The mean fractional refractivity anomalies of JPL in the 20–30 km layer (see Figure 6 and Table 2) decrease from 0.09% to 0.02% in the northern high-latitude zone, and from 0.18% to 0.0% in the southern high-latitude zone (Figure 11). The SD of the time series of fractional refractivity anomalies for JPL and NCAR decreases around 30% in the midlatitudes and to 50% in the high latitudes (Table 2). The SD for JPL in the 20–30 km layer decreases by amounts ranging from 0.4% to 0.17% in the southern high-latitude zone, and decreases from 0.28% to 0.16% in the northern high-latitude zone (Figure 11 and Table 2). Most of the obvious intercenter random differences are sampling error. After sampling errors subtracted, the random retrieval error for each center ranges from 0.02% globally to 0.17% in the high-latitude stratosphere (the SD in Table 2). Although the mean fractional refractivity anomalies of WegC is improved (Table 2), the SD for WegC is actually slightly increased in the northern high-latitude zone. This may be owing to the residual sampling errors resulting from the limited temporal (6 hour bin) and spatial resolution (1°) of NCEP (see below for more details).

[68] 2. To quantify the remaining trend uncertainty among data sets after minimizing their sampling errors, we construct the deseasonalized fractional refractivity anomalies of GFZ, JPL, UCAR and WegC using the new MMCs for each vertical layer and latitudinal zone. The trend difference of each center to the mean from four centers, and the RMS of the new deseasonalized fractional

refractivity anomalies for each center at each latitudinal zone and vertical layer are summarized in Table 3 and Table 4, respectively. Figure 12 depicts the differences between the new deseasonalized fractional refractivity anomalies for each center from the mean of all four centers at the southern high-latitude zone, the tropical zone, and the northern midlatitude zone for the 12–20 km layer (left panels) and the 20–30 km layer (right panels), respectively. Comparing with the corresponding panels in Figure 9, the variation of deseasonalized fractional refractivity anomalies among four centers are similar in the tropics, but decrease dramatically in the middle and high latitudes, especially for those of JPL. The mean RMS for JPL decreases from 0.1% to 0.06%, and from 0.28% to 0.12% in the southern high-latitude zone for the 12–20 km layer and the 20–30 km layer (Table 4), respectively. The RMS for JPL also decreases by about one half in northern midlatitudes in all altitudes.

[69] 3. The smaller RMS of deseasonalized fractional refractivity anomaly difference from the mean among the four centers leads to even smaller trend differences than those in the previous section. The global uncertainty of the trend from four centers in the 8–30 km layer to the mean of four centers for the global region decreases from within $\pm 0.04\%/5$ years to $\pm 0.03\%/5$ years (Table 3). The trend differences for each center to the mean trends all decrease except for that of UCAR and WegC in the 20–30 km layer in the high-latitude zones.

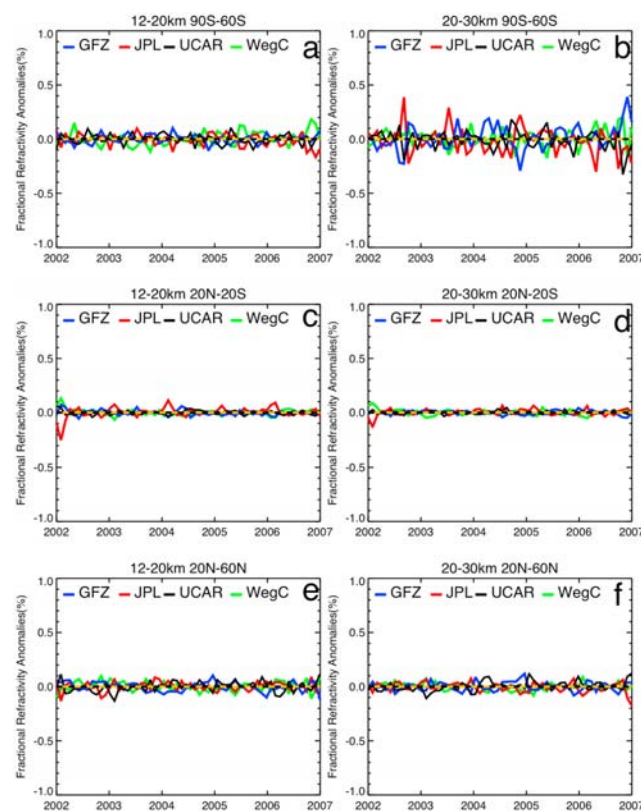


Figure 12. Same as Figure 9 (left 12–20 km layer, right 20–30 km layer, selected latitude bands), but showing the time series of refractivity anomalies with the sampling error subtracted.

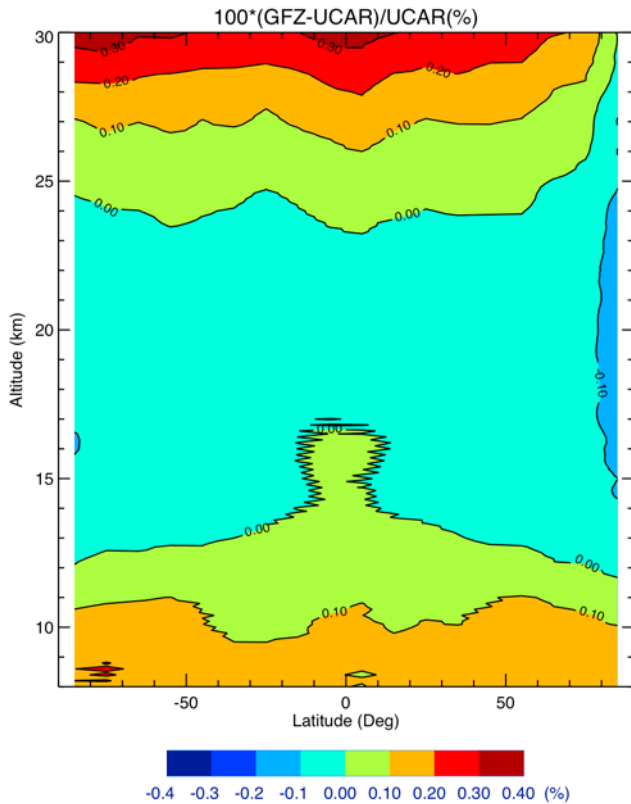


Figure 13. Difference in the mean zonal average refractivity for profile-to-profile UCAR and GFZ fractional refractivity pairs over the period from January 2006 to December 2006.

[70] Note that estimates of sampling error may be limited by the temporal (6 hour bin) and spatial resolution (1°) of NCEP reanalysis especially at the high altitudes over the polar regions, which may lead to residual sampling errors related to sampling differences among the four centers. Using the profile-to-profile UCAR and GFZ comparison (e.g., no sampling differences) in the whole year of 2006, in order to get some additional insight into potential residual sampling effects, we also examined the fractional refractivity anomalies between the two centers (for both these centers individual profiles were available in addition to MMCs). We found that the mean profile-to-profile refractivity difference between UCAR and GFZ is less than 0.05% (Figure 13) in the height between 12 km and 25 km, with a MAD less than 0.2% (not shown). This indicates that the remaining fractional refractivity differences in Figure 10 (e.g., for UCAR in the midlatitudes, and for JPL/WegC at the high latitudes) are still related to the sampling differences between the four centers. Under the condition that there are no sampling differences, no obvious nonzero time series of fractional refractivity anomalies between the two centers in the 8–25 km layer are found (not shown).

5.2. The Uncertainty of Interannual Variability in the Anomalies Among Centers

[71] Although the residual sampling error for the MMC comparison is small, it may still affect the interannual variability in the anomalies among centers. Latitudinal

zones and vertical layers with larger trend differences ($>0.05\%/5$ years, Table 3) to the mean trend are usually related to regions with greater differences in fractional refractivity among four centers ($\text{RMS} > 0.1\%$), which are mainly at the 20–30 km layer in the high-latitude zones, where on average less than 30 CHAMP profiles contribute to the monthly mean. Besides polar regions, most of the mean trend differences at all vertical layers are less than $0.03\%/5$ years.

[72] Because retrieval error contributes only random error, dependent only on the density of soundings, coverage obtained by other GPS RO missions that is denser than that of CHAMP ought to further reduce the contribution of retrieval error (and sampling error) to trend estimates. On average, CHAMP obtains approximately 200 soundings a day over the entire globe. FORMOSAT-3/Constellation Observing System for Meteorology, Ionosphere, and Climate (COSMIC), consisting of six satellites that obtain rising and setting occultations, provides approximately 2000 soundings a day. If these COSMIC soundings yield relatively uncorrelated atmospheric information in most of geographical bins, the error in trend estimates contributed by retrieval error (and sampling error) ought to be reduced by a factor of $\sqrt{2000/200} \approx 3$ over that of CHAMP data.

[73] It is expected that statistically significant trends in refractivity will be attained in ten years [Foelsche et al., 2008b; Leroy et al., 2006a; Ringer and Healy, 2008]. We use a simple scaling law to extrapolate the contribution of retrieval error to climate trends in a 10 year time series of occultation data. For a continuous time series, the serially uncorrelated retrieval error will contribute to uncertainty in a trend estimate proportionally to $(\Delta t)^{-3/2}$, as the work of Leroy et al. [2008, equation (10)]. Scaling the proportionality to the upper bound estimate $0.05\%/5\text{years}$ for a 5 year time series, we find that retrieval error contributes to uncertainty in trends over a time baseline of Δt according to

$$\sigma_{\text{trend, retrieval error}} = \frac{0.1\%}{\Delta t} \times \left(\frac{12 \times (1 \text{ month})}{\Delta t} \right)^{1/2}. \quad (8)$$

Using this scaling law, we deduce that the contribution of retrieval error to a 10 year time series of continuous data, with just one occultation satellite, is $\sim 0.01\%/10$ years. In the final analysis, the fluctuations of natural variability will also contribute to and in fact dominate the uncertainty due to retrieval error. Natural variability is in fact serially correlated and thus requires significance tests such as that of Weatherhead et al. [1998] and Bretherton et al. [1999].

5.3. The Uncertainty of Anomaly Trends due to Inversion Algorithm Among Centers

[74] In the present anomaly trend analysis we demonstrate that, even though there might not be a statistically significant climate trend because of the presence of natural variability, trends derived from GPS RO data are robust for different processing implementations because GPS RO is traceable to international standards. Because trends deduced by various centers are in general within $\pm 0.03\%/5$ years globally, then when a statistically significant climate trend emerges in this data type, we can be confident that all processing centers will agree in the trend estimation. Large

refractivity differences that occur in high-latitude zones in the 20–30 km layer are most likely due to optimal estimation and initialization of the Abel integral used by the four centers. A series of simulation studies would be required to examine how the specific excess phase processing and inversion procedure used by different centers would affect the retrieved refractivity profile. We leave that for a future study.

[75] We note that although removing the sampling errors decreases the RMS differences among the four centers, the absolute mean trend differences are in general within $\pm 0.03\%/5$ years for most of the layers at all latitudinal zones, except for the 20–30 km layers in the polar regions (Tables 3 and 4). The fact that the mean trends of all four centers are similar to those after removing their sampling errors (though the intercenter differences were significantly reduced), except for the northern midlatitudes in the 12–20 km layer and in the southern high latitudes in the 20–30 km layer (see Table 3), implies that although CHAMP provides only a limited sample number of occultation profiles in each 5° latitudinal bin, those single-satellite climatologies are sufficient to construct consistent trend analysis.

6. Conclusions and Outlook

[76] GPS RO is the first satellite remote sensing technique where its fundamental observable can be traced to ultrastable international standards (atomic clocks) on the ground. While the fundamental observable of limb path delay is traceable to the international standard of time, one of the retrieved variables of GPS RO, refractivity, is not. Here we investigate the claim that a retrieval system uniformly applied to the SI traceable observations can yield credible trends in the retrieved GPS RO refractivity. To investigate this claim, we intercompare CHAMP refractivity climate records in the period of 2002–2006 as produced by four essentially independent data centers. We reach the following conclusions.

[77] 1. Despite different approaches and implementations, and the use of differing sets of profiles in the formation of monthly means, the randomly occurring fractional refractivity anomalies among four centers for the MMC differ by less than 0.2% at all latitudes below 25 km. Differences among the centers do introduce nonnegligible systematic refractivity differences at different heights and different latitudes. The largest random refractivity anomaly differences are found near the high-latitude zones above 25 km, which are most likely due to the sampling errors caused by different QC schemes. The mean fractional refractivity anomalies for GFZ, JPL, UCAR and WegC from 8 km to 30 km and from 90°N to 90°S are equal to 0.04%, -0.01% , 0.04%, and -0.06% , respectively. An anomaly for each center indicates the difference from the intermodel mean. The MADs are less than 0.1% for the four centers between 50°S to 50°N , and are within 0.2% in the 50°S to 90°S zone and the 50°N to 90°N zone. Results here demonstrate extreme similarity of the RO data among the four centers, which gives confidence to the feasibility and robustness of the processing chain of each center.

[78] 2. Even with nonzero fractional refractivity anomalies among four centers, because GPS RO data are of very high precision ($\sim 0.1\%$) [e.g., Anthes *et al.*, 2008; Foelsche

et al., 2009; Ho *et al.*, 2009a], the time series of fractional refractivity anomalies for each center are in general highly similar to each other especially in the tropics, and in the midlatitudes. The variation of the time series of fractional refractivity anomalies is shown by their standard deviations varying in different latitudinal zones. In the tropics, the SDs for GFZ, UCAR and WegC are all less than 0.03%, where the SD for JPL is around 0.05%. In the midlatitude zones, the SDs are around 0.08%, where in the high-latitude zones the SDs range around 0.1 to 0.4% for different centers.

[79] 3. We have estimated the uncertainty in refractivity trends induced by retrieval error with the 5 year time series of CHAMP data. The trends estimated by four centers in the 8–12 km layer, 12–20 km layer, and 20–30 km layer and in different latitudinal zones agree within ± 0.05 (%/5 years) globally. The standard deviation in the trend from the four centers in the 12–20 km layer to the mean trend from the four centers for the global zone, northern high-latitude zone, northern midlatitude zone, the tropics, southern midlatitude zone, and southern high-latitude zone are within ± 0.04 , ± 0.03 , ± 0.05 , ± 0.02 , ± 0.06 , and ± 0.04 (%/5 years), and the standard deviation in the trend from the four centers in the 20–30 km layer in the same latitudinal zones are ± 0.04 , ± 0.1 , ± 0.04 , ± 0.01 , ± 0.05 , and ± 0.06 (%/5 years). Note that trends evaluated here should not be considered indicative of any long-term climate trend as the studied period is too short. Results of the anomaly trend analysis show that even a short-term trend derived from GPS RO data is robust with regard to different processing implementations. If the robustness is true for short-term trends, then it can be expected true for long-term trends as well.

[80] 4. The dominant contributor to retrieval differences among the four centers is quality control. The different implementations of quality control cause different subsets of the CHAMP GPS RO data to be retained for the formation of monthly mean climatologies. The gaps in spatial-temporal sampling created by data elimination will consequently be different for the different centers leading to sampling error. When the monthly mean climatologies are corrected for sampling error with the assistance of atmospheric analyses, errors in fractional refractivity trends are reduced from $\pm 0.04\%/5$ years to $\pm 0.03\%/5$ years. This can be considered an upper bound in the processing scheme–induced uncertainty for global refractivity trend monitoring.

[81] 5. All intermodel differences we have uncovered are serially uncorrelated. Thus, the longer the time series, the less significant retrieval error becomes. With a 10 year time series, we expect retrieval errors to contribute no more than 0.01% error in estimated trends in refractivity. Likewise, an increase in sounding density made available with more LEO sounders (i.e., COSMIC) should reduce the error due to sampling and retrieval by the square root of the number of soundings per day by the entire constellation. This should easily satisfy requirements for observing trends in refractivity for climate study ($\approx 0.3\%/decade$) [Leroy *et al.*, 2006b].

[82] We note that the analysis performed here does not recognize retrieval differences that may be systematic to all data centers' processing chains. For example, all data centers have implemented a very similar algorithm in removing the ionospheric influence in observed limb path delay in order to derive an “atmosphere only” refractivity. This algorithm of “ionospheric correction” is not perfect,

and thus there is a retrieval uncertainty associated with it that is commonly called “ionospheric residual” in GPS RO. Because the ionospheric residual will be largely common to all the data centers’ processing presented here, it is not considered in the retrieval difference estimates presented above. The related residual uncertainty at <25 km is expected to be (close to) negligible, since the ionospheric signal is largely dominated by the neutral atmospheric signal at these low heights [e.g., Kursinski *et al.*, 1997; Gobiet and Kirchengast, 2004]. Another such example is the error associated with GPS RO signals that are reflected off the receiver’s low Earth orbiting spacecraft, a retrieval uncertainty commonly called “local multipath.” This retrieval uncertainty enters into the limb path phase, well before GPS RO retrieval chains are applied. Thus, it is not reflected in the retrieval difference estimates presented above. Local multipath uncertainties, though, should be reflected in interspacecraft comparison of GPS RO data products and these as well show excellent consistency [e.g., Foelsche *et al.*, 2009; Ho *et al.*, 2009a].

[83] We expect that improved understanding of the impact of different approaches and implementations among the centers, and using a common set of profiles in the MMCs, will further reduce the intercenter differences. The GPS RO technique, by virtue of its traceability to the international definition of the second, is insensitive to retrieval error when used to estimate inter-annual trends in the climate system.

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